N0TH1NG AI WORKSTATION | SYSTEM ARCHITECTURE v1.8.0

n0th1ng AI

Complete Frontend & Backend Engineering Manual

Exhaustive Technical Specification, Workflows, Protocols, Schema & Codebase Reference

FRONTEND ARCHITECTURE (13 MODULES)

- Zero-Framework Vanilla HTML5 + CSS3 + ES6+ MPA Architecture
- Obsidian & Liquid Gold Design System + 3-Tier Glassmorphism
- Cross-Document View Transitions API + Hover Link Prefetching
- MutationObserver Glassmorphic Select & Ancestor Z-Index Engine
- Chatspace, Files, Image Studio, FramesX, Voice 3D, Forge, ForgeX
- Persona Studio, Telemetry Monitor, Cluster Settings, Robotics

BACKEND CORE & AI ORCHESTRATION

- Node.js + Hono + tRPC v11 (SuperJSON) + Drizzle ORM (34 Tables)
- 5-Stage Tool Router Cascade (lfm2.5-thinking:1.2b + Regex Clusters)
- 25-Round Extended Thinking Agent Loop + Real-Time FenceWatcher
- PhaseTrackerService Multi-Phase Plan Completeness Gate
- Hybrid Semantic + Keyword RAG (nomic-embed-text) + Rolling Memory
- 18 Native Agent Tools (Tavily, Firecrawl, Vision, SQL, Memory)

PYTHON RUNTIMES & MEDIA PIPELINES

- Python Tool Runtime (:8002): Surya OCR/Layout, Marker PDF, Qwen-VL
- 5-Node LangGraph Document Analysis Streaming StateGraph
- ComfyUI (:8188): Flux.1/2 Hires-Fix & LTX-Video 3-Stage Director
- Speech Runtime (:9000): Kokoro, Whisper, Piper, XTTS, Fish, Dia
- Sentence-Pipelined 2-Worker Voice Chat + RIFF WAV Stitching
- NVIDIA PersonaPlex Moshi (:8998) Full-Duplex Speech-to-Speech

AUTONOMOUS CODING, CLUSTER & IOT

- The Forge: THINK -> ACT -> EVALUATE Loop + DAG Contract Tree
- ForgeX: Headless Claude Code CLI Bridge Redirected to Local Ollama
- Distributed GPU Worker Nodes (:3001) + 5s Capabilities Poller
- Cloudflare Tunnel Exposure Gateway (:3443) + Sliding Rate Limiter
- Base64 Cross-Network File Marshaling & Worker Crash Supervisors
- ESP32 Companion Hardware Server (:3005) REST + Binary Audio WS

REPOSITORY & ENVIRONMENT SPECIFICATION

Root Path: d:/AI/Chatbot/app | Worker Path: d:/AI/Chatbot/worker | Version: 1.8.0

Total Document Length: 55 Pages | Coverage: 100% Frontend + Backend + Database + Runtimes

PART I — SYSTEM OVERVIEW & TABLE OF CONTENTS

Master Architecture & Document Roadmap

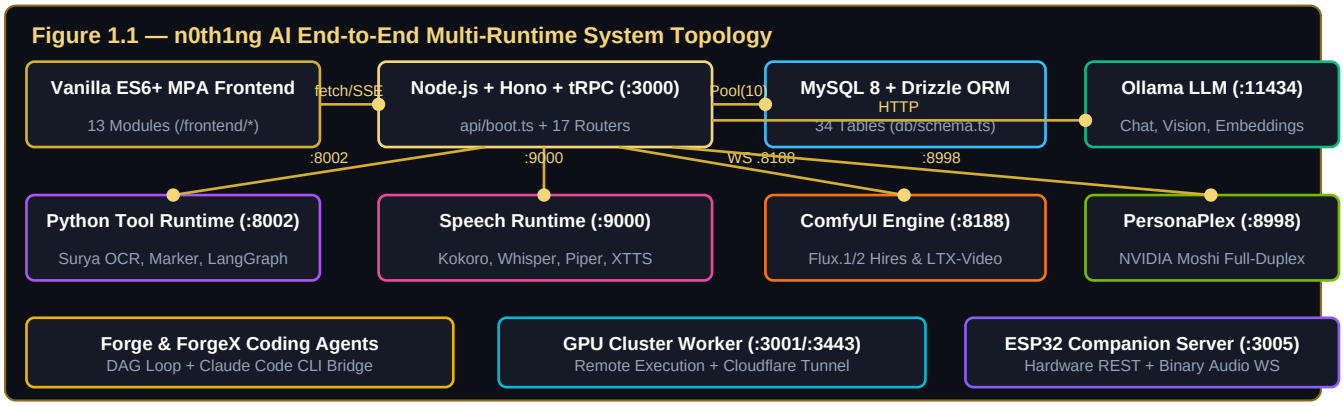
Complete Index of Frontend Modules, Backend Services, Database Schema & Runtimes

1. Document Structure & Engineering Scope

This engineering manual provides an exhaustive, code-verified technical reference for the entire **n0th1ng AI Workstation** repository (**d:/AI/Chatbot/app** and **d:/AI/Chatbot/worker**, version 1.8.0). Every workflow, state machine, wire protocol, database table, DOM controller, streaming parser, and AI microservice is documented down to exact file paths, line ranges, function signatures, and payload contracts.

Part	Section Title	Technical Coverage Summary
Part I	Executive Topology, Tech Stack & Build Pipeline	Architecture reality (Vanilla MPA vs. legacy React scaffold), Vite + Hono SSR dev server, production esbuild bundle, multi-runtime port map.
Part II	Frontend Infrastructure, Design System & Shared Engines	Obsidian & Liquid Gold CSS tokens, 3-tier glassmorphism, fractal SVG noise, nav.js (400ms hover-intent), page-transitions.js (View Transitions + prefetch), liquid-select.js (MutationObserver + z-index climbing), desktop-only.js (hardware screen guard).
Part III	Frontend Modules Deep-Dive (All 13 Workspaces)	Complete state models, DOM hierarchies, user/data workflows, streaming parsers, and function dictionaries for Home, Chatspace, Files, Studio, FramesX, Voice, Persona Studio, Forge, ForgeX, Monitor, Settings, Model Manager, and Robotics.
Part IV	Frontend-to-Backend Wire Protocol & API Matrix	SuperJSON tRPC GET/POST envelope, NDJSON ReadableStream parsing, EventSource SSE named events, multipart FormData uploads, and polling fallbacks.
Part V	Backend Core, Boot Lifecycle & 17 tRPC Routers	api/boot.ts startup sequence, auto-DDL, seeding, embedding backfill, direct Hono REST/SSE endpoints, and every procedure across all 17 tRPC sub-routers.
Part VI	Core AI Orchestration: Chat, RAG, Memory & Agent Loop	Temporal/Memory/Rolling-Summary context builder, Hybrid 0.8*semantic + 0.2*keyword RAG, 5-Stage Tool Router Cascade, 25-round Agent Loop, FenceWatcher, 3-layer recovery cascade, PhaseTrackerService, and all 18 Agent Tools.
Part VII	Specialized AI Runtimes, Media, Coding Agents & Cluster	Python Tool Runtime (:8002) & 5-node LangGraph pipeline, ComfyUI (:8188) 2-pass Hires Image Studio & FramesX 3-stage LTX-Video Director, Speech Runtime (:9000) & PersonaPlex (:8998), The Forge & ForgeX, GPU Worker (:3001/:3443), and ESP32 Companion (:3005).
Part VIII	Automated Source-Verified Schema & Codebase Reference	Complete 34-table column-by-column Drizzle/MySQL dictionary extracted from db/schema.ts and complete file/symbol inventory across the entire codebase.

Figure 1.1 — n0th1ng AI End-to-End Multi-Runtime System Topology



2. Architectural Reality: Vanilla ES6+ MPA vs. Legacy React Scaffold

CRITICAL ARCHITECTURAL NOTE: Inspecting **d:/AI/Chatbot/app/package.json** (#L19-L91) reveals dependencies from an earlier React 19 + Shadcn UI + Tailwind CSS v3 + tRPC React-Query scaffold (such as *react@^19.2.0*, *@radix-ui/**, *@tanstack/react-query*, *zustand*, *three*, *gsap*, and *lucide-react*), alongside **d:/AI/Chatbot/app/index-old.html** (#L1-L13) which references a removed */src/main.tsx* entry point. **None of those React/Tailwind packages are imported or used by the active production frontend.** Instead, the entire live frontend is a **zero-framework, Vanilla HTML5 + CSS3 + ES6+ JavaScript Multi-Page Application (MPA)** served from **d:/AI/Chatbot/app/index.html** and **d:/AI/Chatbot/app/frontend/***.

Subsystem Layer	Technology / Package	Implementation & File Reference
Frontend Architecture	Vanilla HTML5 + CSS3 + ES6+ MPA	13 self-contained module directories under d:/AI/Chatbot/app/frontend/* plus root index.html.
Dev Server & Routing	Vite ^7.2.4 + @hono/vite-dev-server ^0.19.0	d:/AI/Chatbot/app/vite.config.ts (#L9-L15) mounts api/boot.ts on port 3000 with exclude: [/^\/(?!api\/).*$/] so Vite serves static HTML/CSS/JS while Hono handles /api/*.
Production Bundler	vite build + esbuild ^0.27.2	package.json (#L8) outputs static frontend to dist/public and bundles api/boot.ts into dist/boot.js (ESM with createRequire banner).
Production Static Server	@hono/node-server/serve-static	d:/AI/Chatbot/app/api/lib/vite.ts (#L6-L23) serves ./dist/public and falls back to dist/public/index.html for text/html requests.
Frontend Markdown & Code	marked.js + highlight.js v11.11.1	Loaded via CDN in d:/AI/Chatbot/app/frontend/chatspace/index.html (#L11-L13) with github-dark.min.css theme.
Typography	Google Fonts CDN	Inter Tight (weights 100-600), Inter (300-500), and JetBrains Mono (300-500) loaded across all pages.
Primary Backend Server	Hono ^4.8.3 + tRPC ^11.8.1 + SuperJSON	d:/AI/Chatbot/app/api/boot.ts (#L1-L2630) and api/router.ts (#L1-L47) with Zod ^4.3.5 validation.
Database & ORM	MySQL 2 ^3.14.1 + Drizzle ORM ^0.45.1	34 tables in d:/AI/Chatbot/app/db/schema.ts (#L1-L872), pooled in api/queries/connection.ts (#L1-L68).
Python Tool Service	FastAPI + Uvicorn (127.0.0.1:8002)	d:/AI/Chatbot/app/python-runtime/app/main.py (#L1-L79) — Surya OCR/Layout, Marker PDF, Vision, 5-node LangGraph.
Python Speech Service	FastAPI + Uvicorn (127.0.0.1:9000)	d:/AI/Chatbot/app/speech-runtime/main.py (#L1-L41) — Kokoro, Whisper, Piper + FishSpeech (:6101) & Chatterbox (:6102).
Full-Duplex Speech Server	NVIDIA PersonaPlex / Moshi (:8998)	d:/AI/Chatbot/app/api/routers/personaplex.ts (#L1-L400) — nvidia/personaplex-7b-v1 over HTTPS/WSS.
Media Generation Engine	ComfyUI HTTP + WebSocket (:8188)	d:/AI/Chatbot/app/api/services/comfy.ts (#L1-L626) + framesx.ts (#L1-L1036) + FFmpeg scene conditioning.
Distributed Worker Node	Hono (:3001) + Cloudflare Tunnel (:3443)	d:/AI/Chatbot/worker/src/server.ts (#L1-L413) & exposureService.ts (#L1-L663) + Electron Tray.
IoT Companion Server	Node HTTP + ws WebSocketServer (:3005)	d:/AI/Chatbot/app/api/services/companion.ts (#L1-L820) — ESP32 binary audio + Whisper + Gemma3 + Kokoro.

PART II — FRONTEND INFRASTRUCTURE & DESIGN SYSTEM

Obsidian & Liquid Gold Styling & Shared DOM Engines

Deep Analysis of frontend/assets/*, CSS Custom Properties, Glassmorphism & View Transitions

1. The 'Obsidian & Liquid Gold' Design System & CSS Architecture

Every module stylesheet (e.g., **d:/AI/Chatbot/app/frontend/home/styles.css** #L1-L65, **d:/AI/Chatbot/app/frontend/chatspace/chat.css** #L1-L27) implements a unified dark-first token hierarchy built on CSS Custom Properties, 3-tier *backdrop-filter* glassmorphism, and an inline SVG fractal noise overlay:

- **Core Surface & Text Tokens**: *--bg: #030303* (*#050505* in Voice Studio), *--surface: #0a0a0a*, *--text: #F5F5F7*, *--text-secondary: #86868B*, *--text-muted: #52525B*.
- **Liquid Gold Brand Tokens**: *--gold: #D4AF37*, *--gold-bright: #F5D77A*, *--gold-deep: #8A6A1F*, and *--grad-aurora: linear-gradient(135deg, #8A6A1F 0%, #D4AF37 40%, #F5D77A 70%, #F0E2B6 100%)*.
- **Module Theme Overrides**:
  - **Persona Studio (personastudio.css #L9-L26)**: Overrides accents to NVIDIA Green (*--nvidia-green: #76b900*, *.glass-nvidia*).
  - **Voice Studio (voice.css #L9, #L44-L90 & voice.js #L755)**: Uses a live *--theme-color-rgb* variable updated at 60fps by *Visualizer.animate()* to smoothly interpolate all borders, shadows, and text glows across 6 state-driven RGB palettes (Cyan, Purple, Crimson, Gold, Emerald, Orange).
- **3-Tier Glassmorphism & Fractal Noise**:
  - *.glass*: *background: rgba(255,255,255,0.025); backdrop-filter: blur(30px) saturate(120%); border: 1px solid rgba(255,255,255,0.07)*.
  - *.glass-strong*: *background: rgba(12,12,14,0.78); backdrop-filter: blur(50px) saturate(140%)*.
  - *.glass-liquid*: Adds directional specular top/left border highlights and *linear-gradient(135deg, rgba(255,255,255,0.05) 0%, rgba(255,255,255,0.01) 100%)*.
  - *.noise::after* (**home/styles.css #L60-L65**): Renders a fixed fullscreen data-URI SVG *<feTurbulence type='fractalNoise' baseFrequency='0.9' numOctaves='2'/>* at *opacity: 0.03* and *mix-blend-mode: overlay* to eliminate gradient banding.

2. Shared Runtime Engines in d:/AI/Chatbot/app/frontend/assets/

2.1 Global Navbar & 400ms Hover-Intent Mega-Menu (nav.js #L1-L54 & nav.css #L1-L80)

- **Scroll Compression (#L27-L37)**: Passive scroll listener toggles *.scrolled* on *#navWrap* whenever *window.scrollY > 20*, tightening padding and increasing glass opacity.
- **Mobile Off-Canvas Drawer (#L6-L24)**: Binds *#burger*, *.drawer-bg*, and *[data-close]* links to toggle *.open* on *#drawer* and lock *document.body.style.overflow*.
- **Hover-Intent Grace Timer (#L41-L53)**: On *#ddFeatures*, *mouseenter* clears any pending timeout and adds *.open*; *mouseleave* starts a **400ms grace timer** (*closeTimer = setTimeout(..., 400)*) before removing *.open* so diagonal cursor travel from the navbar trigger into the multi-column module grid never accidentally collapses the menu.

2.2 Cross-Document View Transitions & Hover Prefetcher (page-transitions.js #L1-L140 & page-transitions.css #L1-L113)

- **Native MPA View Transitions (page-transitions.css #L14-L50, #L108-L109)**: Declares *@view-transition { navigation: auto; }*. Isolates *#navWrap* with *view-transition-name: n0-nav !important* so the navbar remains pinned in place while *::view-transition-old(root)* runs *vt-out* (180ms fade + 0.985 scale) and *::view-transition-new(root)* runs *vt-in* (260ms fade + translateY).
- **Top Gold Progress Bar Lifecycle (page-transitions.js #L20-L104)**: Dynamically creates *#vt-bar*. Calls *showBar()* (#L33-L50) on internal *<a>* clicks (#L80-L94) and *pageswap* (#L68-L70), and calls *completeBar()* (#L53-L62) on *pagereveal* (#L72-L74) and *pageshow* bfcache restores (#L99-L104).
- **Intelligent Desktop Hover Prefetching (#L112-L137)**: On desktop viewports (*matchMedia('(min-width: 901px)')*) when *navigator.connection.saveData* is false and effectiveType is not *2g*, hovering any same-origin *<a href>* records the URL

in a deduplicating *Set* (#L118) and appends *<link rel='prefetch' href='...'>* to *<head>*.

2.3 Reactive Glassmorphic Engine (liquid-select.js #L1-L199 & liquid-select.css #L1-L188)

- **DOM Construction (initLiquidSelect #L30-L198)**: Hides the native *<select>* via *.liquid-select-hidden* and inserts a *.liquid-select-wrapper* containing *.liquid-select-trigger* (displaying selected text + animated chevron SVG) and *.liquid-select-dropdown*.
- **Recursive Option & Optgroup Builder (renderOptions #L66-L135)**: Recursively traverses *<optgroup>* (rendered as *.liquid-select-group-title*) and *<option>* elements. Selecting an item updates *nativeSelect.value* and dispatches a synthetic *new Event('change', { bubbles: true })* (#L118) so module event listeners work transparently.
- **Stacking Context Elevation Fix (toggleElevation #L139-L159)**: Because CSS *backdrop-filter* creates a new stacking context on every glass card, an open dropdown inside a card would normally clip behind sibling cards. When opened, *toggleElevation(true)* walks up the DOM ancestor tree up to *<main>* or *.workspace* and sets *el.style.zIndex = '99999'*, restoring original z-indices on close.
- **Async API Synchronization via MutationObserver (#L189-L192)**: Attaches a *MutationObserver* watching *{ childList: true, subtree: true, attributes: true, attributeFilter: ['selected'] }* on the hidden *<select>*. Whenever an async *fetch()* populates models, voices, or samplers into the native select, the glass dropdown rebuilds automatically.

2.4 Hardware-Level Phone Guard (desktop-only.js #L1-L44 & desktop-only.css #L1-L69)

- Loaded synchronously in *<head>* on **Forge** (*frontend/forge/index.html #L8-L9*) and **ForgeX** (*frontend/forgex/index.html #L8-L9*).
- **Defeating 'Request Desktop Site' Spoofing (isPhoneDevice #L18-L34)**: Mobile browsers in desktop mode spoof *window.innerWidth* (~980px), bypassing CSS media queries. *isPhoneDevice()* inspects physical hardware dimensions via *Math.min(window.screen.width, window.screen.height) <= 540* (*PHONE_MAX_SHORT_SIDE = 540*) combined with *navigator.maxTouchPoints > 0* or *matchMedia('(pointer: coarse)')*. When true, it adds *html.n0-desktop-only* (#L36-L38), activating *#n0DesktopOnlyGuard* (*z-index: 2147483647*) and hiding all other *body > ** elements.

PART III — ALL 13 FRONTEND WORKSPACES IN DETAIL

Complete Workflows, State Machines, Streaming Parsers & DOM Controllers

Line-by-Line Technical Breakdown of Every Module Under d:/AI/Chatbot/app/frontend/*

Module 1: Home Landing & Workstation Portal (index.html & frontend/home/*)

Files: *d:/AI/Chatbot/app/index.html* (#L1-L315), *d:/AI/Chatbot/app/frontend/home/index.html* (#L1-L318), *script.js* (#L1-L160), *styles.css* (#L1-L680).

- **Hero Scroll Parallax (script.js #L5-L16)**: Computes *p = Math.min(1, window.scrollY / 600)* on scroll and applies *transform: translateY(${p * 60}px) scale(${1 - p * 0.06})* and *opacity: ${1 - p * 0.9}* to *.hero-inner*.
- **80-Node Gold Constellation Canvas (particles() #L32-L83)**: Clamps *dpr = Math.min(2, window.devicePixelRatio || 1)* (#L35), spawns 80 gold particles (*rgba(245,215,122,0.75)*) with random velocity vectors, bounces them off viewport edges, and draws O(N²) proximity lines (*rgba(212,175,55, (1 - dist/max) * 0.25)*) whenever two particles come within *140 * dpr* pixels.
- **Live Image Grid Poller (imgGrid() #L86-L114)**: Queries *GET /api/trpc/image.list* on boot and every *10,000ms* (#L113), rendering the 4 newest ComfyUI generated images into *#imgGrid* (falling back to 4 curated Unsplash URLs if the gallery is empty).
- **Sinusoidal Sparkbars & Scroll-Reveal Counters (#L117-L159)**: *spark()* (#L117-L125) populates 28 sinusoidal bars into *#spark*. An *IntersectionObserver* (*threshold: 0.15*, #L129-L136) adds *.in* to *.reveal* sections and triggers *animateCounters()* (#L142-L152), which animates *[data-count]* KPI numbers over *1800ms* using cubic ease-out *1 - Math.pow(1 - p, 3)*.

Module 2: Chatspace — Core LLM Chat, RAG, Multimodal Attachments & Agent Loop (frontend/chatspace/*)

Files: *d:/AI/Chatbot/app/frontend/chatspace/index.html* (#L1-L320), *chat.js* (#L1-L1490), *chat.css* (#L1-L620).

- **Module State (chat.js #L1-L6)**: *currentConversationId* (string|null), *currentModel* (string|null), *selectedKnowledgeFiles* (*Set<string>*), *attachedFiles* (*File[]*), *useRag* (boolean, default true), *agentMode* (boolean, default false).
- **Boot & Cross-Module Voice Transcript Handoff (#L1458-L1488)**:
  1. Checks *localStorage.getItem('voiceTranscript')* (#L1459-L1468) and *sessionStorage.getItem('voiceStudioTransfer')* (#L1477-L1488 — written by Voice Studio's STT 'To Chatspace' button in *voice.js #L906-L912*). If present, pre-fills *#chatInput*, auto-resizes the textarea, and removes the temporary storage key.
  2. Executes *loadModels()* (#L1203-L1238, fetching *GET /api/trpc/model.list* and grouping models into *Open Router Models* and *Ollama Models* <optgroup> elements), *loadConversations()* (#L1279-L1314, *GET /api/trpc/conversation.list*), and *loadKnowledgeFiles()* (#L1240-L1269, *GET /api/trpc/file.list*).
- **Multimodal Attachment Capture (#L1063-L1130)**: Captures files from 3 sources: drag-and-drop onto *.chat-area* (#L1070-L1077), clipboard image/file paste on *#chatInput* (#L1080-L1095), and *#attachBtn* file input (#L1098-L1106). *renderAttachmentChips()* (#L1116-L1130) renders removable chips in *#chatAttachments*.
- **Send Message & Streaming Lifecycle (handleSend() #L200-L276)**:
  1. Validates non-empty *text* or *attachedFiles.length > 0* (#L203).
  2. If *!currentConversationId*, lazily creates a conversation via *createConversation(text.slice(0, 40))* (#L207-L209 -> *POST /api/trpc/conversation.create* #L1135-L1149).
  3. If *attachedFiles.length > 0*, uploads files via *POST /api/chat-files/upload* using *FormData* (*conversationId + files*, #L222-L259), receiving processed attachment metadata.
  4. Renders the user bubble via *appendUserMessage(text, uploadedAttachments)* (#L781-L814) and persists to MySQL via *saveMessage('user', text)* (#L1151-L1169 -> *POST /api/trpc/message.create*).
  5. Branches on *agentMode* (#L269-L275):
- **Branch A — Standard Streaming (generateRealResponse() #L278-L442)**: Sends *POST /api/chat/stream* (#L294-L307) with *{ model, prompt, selectedKnowledgeFiles, uploadedAttachments, conversationId, useRag, timezone:*

Intl.DateTimeFormat().resolvedOptions().timeZone }. Reads NDJSON lines via *response.body.getReader()* + *TextDecoder* (#L333-L414):

  - *data.thinking*: Lazily creates *<details class='thinking-box' open>*, starts a 100ms elapsed timer (*createThinkingTimer()* #L10-L37), and appends tokens to *#thinkingContent* (#L350-L376).
  - *data.status* / *data.error*: Renders inline status text (#L378-L382) or *.error-box* (#L384-L389).
  - *data.response*: Stops the thinking timer, collapses *thinkingBox.open = false*, appends tokens to *aiText*, renders HTML via *marked.parse(aiText)*, and highlights code blocks via *hljs.highlightElement(block)* (#L391-L407).
  - Post-Stream: Persists assistant message via *saveMessage('assistant', aiText, thinkingText)* (#L428) and queries *GET /api/chat/sources?conversationId=...* (#L431-L436) to populate the right-hand RAG citations drawer via *renderSourcesPanel()* (#L943-L968).

- **Branch B — Multi-Round Agentic Tool Loop (generateAgentResponse() #L460-L778)**: Sends *POST /api/chat/agent-stream* (#L483-L496). Tracks multi-round state in *roundsData* and constructs per-round collapsible containers via *startRoundBlock(roundNum)* (#L510-L536):
  - *{ round }*: Finalizes previous round timer and initializes Round N block (#L569-L579).
  - *{ thinking }*: Streams reasoning into Round N's *<pre class='thinking-content'>* (#L583-L591).
  - *{ tool_call }*: Creates a *.tool-chip.running* badge with tool icon (*TOOL_ICONS* #L445-L453), tool name, truncated arguments (*formatToolArgs()* #L455-L458), and a live 100ms execution timer (#L595-L639).
  - *{ tool_result }*: Transitions the matching chip from *.running* to *.done*, records elapsed seconds, and appends an expandable *<details class='tool-chip-details'>* result viewer (#L643-L686).
  - *{ summary_chunk }* / *{ summary_done }*: Streams a concise post-round summary into *.thinking-summary* (#L708-L735).
  - *{ response }*: Streams the final synthesized Markdown answer and persists the entire multi-round trace via *saveExtendedMessage(finalText, roundsData)* (#L766 -> *POST /api/trpc/message.createWithToolCalls* #L1177-L1201).

Module 3: Files & LangGraph Document Intelligence (frontend/files/*)

Files: *d:/AI/Chatbot/app/frontend/files/index.html* (#L1-L260), *files.js* (#L1-L878), *files.css* (#L1-L690).

- **Folder & File CRUD (files.js #L95-L295, #L368-L410)**: Manages folders (*loadFolders* #L95 -> *file.listFolders*, *file.createFolder* #L163, *file.deleteFolder* #L187) and files (*loadFiles* #L201 -> *file.list*, *deleteFile* #L273 -> *file.delete*). Implements a 300ms debounced search on *#fileSearch* (#L368-L410 -> *GET /api/trpc/file.search*) and Grid/List layout toggling (#L55-L67).
- **Upload & Automatic 5-Node LangGraph SSE Stream (#L298-L745)**:
  1. Dropping or selecting files triggers *handleFilesUpload() -> uploadFile(file)* (#L311-L343), posting *FormData* to *POST /api/files/upload*.
  2. When *uploadResult?.fileId* returns, it automatically invokes *startDocumentAnalysis(uploadResult.fileId, file.name)* (#L332-L335).
  3. *startDocumentAnalysis()* (#L591-L699) opens a streaming SSE reader via *fetch('/api/files/${fileId}/analyze-stream')* (#L604) and tracks *PIPELINE_NODES = ['ingestion', 'summarizer', 'extractor', 'metadata', 'synthesizer']* (#L544) inside *activeAnalyses* (Map).
  4. *handleStreamEvent()* (#L646-L699) updates both the bottom-right floating progress toast (*renderToast()* #L550-L589) and the modal stepper (*updateModalProgress()* #L701-L745) on *event: progress*, streams live LLM reasoning tokens into *#modalLiveThinkingText* on *event: thinking* (#L661-L667), and auto-dismisses the toast 4000ms after *event: done* (#L676-L681).
- **3-Tab File Preview Modal (openPreviewModal() #L424-L536 & #L797-L865)**: Tab 1 (*preview*) renders native *<img>*, *<video>*, *<audio>*, PDF *<iframe>*, or text snippet; Tab 2 (*extracted*) lazy-loads OCR/PDF text via *GET /api/trpc/fileContent.getByFileId* (#L470-L530) with copy support; Tab 3 (*analysis*) fetches *GET /api/files/${fileId}/analysis* (#L797-L824) and renders Executive Summary, Key Insights, Q&A; Pairs, Topics & Entities tags, and Document Metadata via *renderAnalysisResults()* (#L826-L865).

Module 4: Image Studio — ComfyUI & 2-Pass Hires Upscaling (frontend/studio/*)

Files: *d:/AI/Chatbot/app/frontend/studio/index.html* (#L1-L420), *studio.js* (#L1-L1341), *studio.css* (#L1-L910).

- **Provider & ComfyUI Node Discovery (studio.js #L1171-L1326)**: *loadImageProviders()* (#L1171-L1227) fetches *providers.list* (*mediaType: 'image'*). Selecting a 2-pass *full-pipeline* provider reveals *#upscaleSettingsGroup* and applies

defaults via *applyUpscaleDefaults()* (#L1229-L1288). *loadComfyOptions()* (#L1293-L1326) queries *providers.comfyOptions* to populate live ComfyUI samplers and upscale models.

- **GCD Aspect Ratio Calculator & Snap-to-8 (#L124-L284)**: Computes greatest common divisor via *gcd(a, b)* (#L169) and maps dimensions to *KNOWN_RATIOS* (#L124-L134) in *formatRatio()* (#L171-L195). When *#lockRatioToggle* is active, changing width or height scales the opposite dimension proportionally and snaps to multiples of 8 (*Math.round(... / 8) * 8*, #L240-L250).
- **SSE Live Progress, Polling Fallback & Split-Pane Comparison (#L395-L644)**:
  1. Clicking *#generateBtn* calls *POST /api/image/generate* (#L417-L433) and connects *new EventSource('/api/image/stream/${jobId}')* (*connectLiveStream* #L448-L547).
  2. Translates ComfyUI node IDs via *STAGE_LABELS* (#L340-L350: Pass 1 base generation -> ESRGAN upscale -> Pass 2 Hires refinement) and updates *#progressBarFill* on *comfy_progress*. If SSE drops, falls back to 1000ms polling via *pollGeneration(jobId)* (#L551-L580).
  3. *loadResult(jobId)* (#L584-L644) inspects *result.baseImageUrl*: if present, activates the side-by-side split viewport (*#canvasSplit* with *#basePane* 'Base Pass 1' and *#finalPane* 'Upscaled Final'); otherwise displays the single image.
- **Lightbox Modal & Parameter Reuse (#L917-L1167)**: *openImageModal(item, index)* (#L943-L979) supports keyboard navigation (*ArrowLeft*, *ArrowRight*, *Escape* #L1149-L1166) and *reuseConfiguration(item)* (#L988-L1081) to restore all prompt, seed, CFG, sampler, and hires parameters into the left sidebar.

Module 5: FramesX — Multi-Scene AI Video Studio (frontend/framesx/*)

Files: *d:/AI/Chatbot/app/frontend/framesx/index.html* (#L1-L390), *framesx.js* (#L1-L790), *framesx.css* (#L1-L960).

- **Snap-to-32 Resolution & Multi-Scene Threshold (framesx.js #L11, #L203-L344)**: Custom dimensions snap to multiples of 32 via *snapTo32()* (#L203-L208). When *#durationSlider* exceeds *SINGLE_SCENE_MAX = 6* seconds (*updateDurationUI()* #L215-L236), *#multiSceneNotice* and LLM scene planning model controls (*loadPlanningModels()* #L250-L289) activate.
- **14-Event Multi-Scene SSE Stream (startGeneration #L365-L685)**: Calls *POST /api/video/generate* and opens *EventSource('/api/video/stream/${jobId}')* (*connectLiveStream* #L500-L598), handling *planning_start*, *planning_token* (streams LLM scene prompt expansion tokens live into *#planningStreamText* #L516-L523), *planning_done* (renders per-scene cards via *renderScenePlan()* #L647-L685), *scene_start*, *comfy_queued*, *comfy_node*, *comfy_progress* (updates overall + per-scene progress bar *#scene-bar-${i}*), *comfy_cached*, *scene_done*, *scene_error*, *merging_start*, *merging_done*, *done* (*loadResult* #L619-L645 loads the merged MP4 into *#resultVideo*), and *error*.

Module 6: Voice Studio — 3D Particle Visualizer & Full-Duplex Speech (frontend/voice/*)

Files: *d:/AI/Chatbot/app/frontend/voice/index.html* (#L1-L340), *voice.js* (#L1-L2226), *voice.css* (#L1-L820).

Architected around **9 Object-Oriented Classes** in *voice.js*:

Class & Line Range	Technical Responsibilities & Internal Mechanics
`VoiceApi (#L47-L153)`	Static REST/tRPC client for /api/speech/* (/health, /providers, /models, /voices, /profiles, /tts, /stt), /api/voice/* (/conversations, /conversation, /message), /api/chat/stream, and /api/trpc/model.list.
`SettingsManager (#L158-L170)`	Holds active ttsProvider, ttsModel, ttsVoice, sttProvider, sttModel, llmModel, speed, pitch, autoPlay, handsFree, inputDeviceId, outputDeviceId.
`DeviceManager (#L175-L200)`	Enumerates hardware microphones and speakers via navigator.mediaDevices.enumerateDevices().
`AudioEngine (#L205-L330)`	Singleton AudioContext + AnalyserNode (fftSize = 1024) + reusable <audio> element via MediaElementAudioSourceNode. Uses a monotonic _playToken counter (#L213, #L246-L315) so rapid segment transitions never cause uncaught AbortError rejections.
`AudioQueue (#L340-L415)`	Out-of-order sequence reassembly buffer (_pending: Map<seq, base64>, _nextSeq) that receives { audioSegment, seq } chunks from /api/chat/stream and plays sentences strictly in sequential order (0, 1, 2...).
`HistoryManager (#L420-L512)`	Manages local TTS/STT history items and DB-backed STS voice conversations (loadConversationsFromDB #L430-L448).

Class & Line Range	Technical Responsibilities & Internal Mechanics
`Visualizer (#L517-L767)`	Renders 500 3D-projected particles on #voiceCanvas. Morphs formations (STAGE_BY_STATE #L34-L42) between SCATTERED (idle), SPHERE (3D Fibonacci sphere pulsing with mic RMS energy), and WAVE (3D sinusoidal ribbon modulated by FFT bins). Smoothly interpolates THEMES RGB arrays (#L20-L28) and writes --theme-color-rgb to document.documentElement.style (#L755) on every frame.
`StateManager (#L772-L778)`	Updates #stateText HUD overlay and synchronizes Visualizer.setState().
`VoiceUI (#L783-L1849) & VoiceConversation (#L1854-L2219)`	Orchestrates Mode 1 (TTS synthesis + voice cloning upload POST /api/speech/profiles), Mode 2 (MediaRecorder audio/webm;codecs=opus -> POST /api/speech/stt + 'To Chatspace' sessionStorage transfer), and Mode 3 (STS Live Voice Chat via webkitSpeechRecognition with 1500ms silence timer -> POST /api/chat/stream responseMode:'voice' -> AudioQueue playback -> POST /api/voice/message persistence). Includes startVoiceListPolling() (#L1160-L1206) polling /api/speech/voices every 4000ms while remote worker runtimes boot.

Module 7: Persona Studio — NVIDIA PersonaPlex Manager (frontend/personastudio/*)

Files: *d:/AI/Chatbot/app/frontend/personastudio/index.html* (#L1-L210), *personastudio.js* (#L1-L371), *personastudio.css* (#L1-L380).

- **Parallel Initialization (init() #L43-L65)**: Runs *Promise.all* across *personaplex.voices*, *personaplex.personas.list*, *personaplex.setup.status*, and *personaplex.server.status*.
- **Setup & Server State Machine (renderServerState() #L105-L213)**: Handles *!setupStatus.installed* (*not_started* -> 'Install PersonaPlex' calling *personaplex.setup.start*; *cloning/installing* -> polls every 2000ms via *startPollingIfNeeded()* #L67-L97 and streams logs into *#activeLogConsole*; *error* -> retry) and *setupStatus.installed* (*stopped* -> collects HuggingFace token *#hfToken* and CPU Offload toggle *#cpuOffload* -> calls *personaplex.server.start* #L241-L256; *starting* -> 2000ms log polling; *running* -> renders Online badge + external link to *serverStatus.url* + 'Stop Server' button). Also provides Persona CRUD (#L270-L347).

Module 8: The Forge — Autonomous Build Agent (frontend/forge/*)

Files: *d:/AI/Chatbot/app/frontend/forge/index.html* (#L1-L240), *forge.js* (#L1-L798), *forge.css* (#L1-L580).

- **Dual Ambient Canvases (forge.js #L18-L145)**: *liquidField()* (#L24-L85) renders 5 pulsing molten-gold radial gradient metaballs with *globalCompositeOperation = 'lighter'*; *particles()* (#L91-L145) renders a 55-node gold particle network.
- **Cross-Chunk Tag Splitter (streamRawText() #L147-L275)**: Maintains a 10-character tail buffer (*TAG_BUFFER_MAX = 10*, #L168) so *<think>* and *</think>* tags split across arbitrary SSE token chunks are cleanly detected, routing reasoning tokens into dim *.raw-think-text* spans (*appendThinkText* #L257-L265) and action JSON into bright *.raw-final-text* spans (*appendFinalText* #L267-L275).
- **Task Tree & 5-Event SSE Stream (connectLiveLog #L456-L625)**: *renderTaskTree()* (#L456-L492) computes task depth via *depthOf()* (#L464-L469) and wires *focusTask(taskId)* (#L494-L506 -> *forge.getTaskLog*). *EventSource('/api/forge/${sessionId}/stream')* listens to *iteration* (renders *[THINK]/[ACT]/[EVALUATE]* groups), *token*, *status*, *raw_activity* (inserts dashed run separators on *false -> true* transitions), and *error*. *sendFollowUp()* (#L768-L798) appends new tasks when status is *done*, *blocked*, or *paused*.

Module 9: ForgeX — Local Claude Code CLI Bridge (frontend/forgex/*)

Files: *d:/AI/Chatbot/app/frontend/forgex/index.html* (#L1-L210), *forgex.js* (#L1-L360), *forgex.css* (#L1-L520).

- **Workflow (forgex.js #L98-L360)**: *FORGEX_API* (#L98-L127) wraps */api/trpc/forgex.** (*list*, *listModels*, *create*, *sendInput*, *stop*). *connectLiveTerminal(sessionId)* (#L237-L266) subscribes to *EventSource('/api/forgex/${sessionId}/stream')* for *output* (*stdout/stderr/[SYSTEM]* lines in *#forgexTerminal*) and *status*. Because stopping a session kills only the OS child process while keeping *claudeSessionId* in MySQL, *updateFollowupBar()* (#L213-L224) enables resumable follow-ups (*claude -p --resume*) whenever status is *idle* or *exited*.

Module 10: System Telemetry Monitor (frontend/monitor/*)

Files: *d:/AI/Chatbot/app/frontend/monitor/index.html* (#L1-L280), *monitor.js* (#L1-L323), *monitor.css* (#L1-L430).

- **4-Rate Concurrent Polling Architecture (monitor.js #L27-L323)**: Runs 4 independent intervals: (1) **1000ms** *updateSystemStats()* (#L66-L110 -> *GET /api/system/current*) updating CPU%, GPU%, RAM%, VRAM%, CPU/GPU SVG circular *stroke-dasharray* thermal gauges, and network KB/s; (2) **3000ms** *updateProcesses()* (#L188-L230 -> *GET /api/system/processes*); (3) **5000ms** *updateServices()* (#L138-L184 -> *GET /api/system/services*); (4) **15000ms** *updateStorage()* (#L234-L282 -> *GET /api/system/storage*), plus a 2.5s event feed capped at 8 DOM nodes (#L297-L299).

Module 11: Settings & Distributed Cluster Control (frontend/settings/*)

Files: *d:/AI/Chatbot/app/frontend/settings/index.html* (#L1-L540), *settings.js* (#L1-L562), *settings.css* (#L1-L490).

- **Lazy Tab Loaders (settings.js #L97-L364)**: *sectionLoaders* (#L97-L103) lazily fetches data on first tab visit: *sec-models* (*loadModels* #L165-L236 -> *model.list*, *model.load* with *keepAlive: -1*, *model.unload*), *sec-python* (*loadPython* #L239-L268 -> *runtime.python.{status,start,stop}*), *sec-speech* (*loadSpeech* #L271-L324 -> *speech.runtime.{status,start,stop}*), and *sec-cluster* (*loadCluster* #L327-L364 -> *cluster.workers*).
- **Cluster Worker Telemetry & SSE Log Modal (openWorkerModal #L366-L562)**: Polls *cluster.metricHistory* every *4000ms* (#L374, *refreshWorkerMetrics* #L432-L460), draws 7 dependency-free 2D <*canvas*> sparklines via *drawSparkline()* (#L465-L503: CPU, RAM, GPU, CPU Temp, GPU Temp, Net RX, Net TX), and streams live worker/Kokoro stdout via *new EventSource('/api/cluster/${workerId}/logs/stream?source=...')* (*startLogStream* #L506-L546, capped at 500 DOM nodes #L559-L561).

Module 12: Model Manager (frontend/model/*) & Module 13: Robotics (frontend/robotics/*)

- **Model Manager (model-manager.js #L1-L426)**: 8-class Pub/Sub architecture: *EventManager* (#L9-L35), *ApiManager* (#L51-L118), *StateManager* (#L123-L152), *ComponentFactory* (#L157-L230, converting camelCase capabilities into Title Case badges), *FilterManager* (#L236-L274), *OverviewManager* (#L276-L291), *PackageManager* (#L293-L317), *ActionManager* (#L319-L348), and *DownloadManager* (#L350-L398, animating *#download-dock-root*).
- **Robotics Dashboard (robotics.js #L1-L81)**: *updateTopologyLines()* (#L27-L47) computes bounding-box center coordinates on resize to connect *#n-core* to 4 peripheral nodes (*ESP-ALPHA*, *ESP-BETA*, *CAM-01*, *SENS-HUB*) via animated dashed SVG <*path class='t-line'*> elements, paired with a 2500ms environmental sensor loop (#L58-L81).

Frontend Algorithmic & Mathematical Deep-Dives

1. Voice Studio 3D Particle Projection & Audio-Reactive Morphing (voice.js #L517-L767)

```
// 1. Fibonacci Sphere Formation (SPHERE stage — listening / thinking / processing)
const phi = Math.acos(1 - 2 * (i + 0.5) / this.particleCount);
const theta = Math.PI * (1 + Math.sqrt(5)) * i;
const r = baseRadius * (1 + rmsEnergy * 0.65);
targetX = r * Math.sin(phi) * Math.cos(theta);
targetY = r * Math.sin(phi) * Math.sin(theta);
targetZ = r * Math.cos(phi);

// 2. 3D Perspective Projection + 60fps RGB Theme Interpolation
const fov = 450;
const scale = fov / (fov + rotatedZ + 500);
const screenX = centerX + rotatedX * scale;
const screenY = centerY + rotatedY * scale;
this.currentRgb[c] += (this.targetRgb[c] - this.currentRgb[c]) * 0.05;
document.documentElement.style.setProperty('--theme-color-rgb', rgbString);
```

2. Forge Cross-Chunk Tag Splitter State Machine (forge.js #L165-L275)

```
// Handles <think> and </think> tags split across arbitrary SSE token boundaries
const TAG_BUFFER_MAX = 10; // length of '</think>' is 8 chars
rawTagBuffer += incomingChunk;
while (rawTagBuffer.length > 0) {
  const targetTag = rawInThinkBlock ? '</think>' : '<think>';
  const idx = rawTagBuffer.indexOf(targetTag);
  if (idx !== -1) {
    const before = rawTagBuffer.slice(0, idx);
    if (before) (rawInThinkBlock ? appendThinkText : appendFinalText)(before);
    rawInThinkBlock = !rawInThinkBlock;
    rawTagBuffer = rawTagBuffer.slice(idx + targetTag.length);
  } else if (rawTagBuffer.length > TAG_BUFFER_MAX) {
    const flushLen = rawTagBuffer.length - TAG_BUFFER_MAX;
    (rawInThinkBlock ? appendThinkText : appendFinalText)(rawTagBuffer.slice(0, flushLen));
    rawTagBuffer = rawTagBuffer.slice(flushLen);
    break;
  } else { break; }
}
```

3. Complete 13-Module DOM ID, State & Storage Key Reference Matrix

Module & Controller	Key JS State Variables	Primary DOM Element IDs	Storage / Persistence
Home (home/script.js)	window.scrollY, IntersectionObserver	#particles, #imgGrid, #spark, #stats, .hero-inner, .reveal, [data-count]	None (stateless portal)
Chatspace (chatspace/chat.js)	currentConversationId, currentModel, selectedKnowledgeFiles (Set), attachedFiles, useRag, agentMode	#chatStream, #chatInput, #sendBtn, #modelSelect, #extendedThinkingToggle, #ragToggle, #attachBtn, #chatAttachments, #conversationList, #knowledgeFileList, #sourcesPanel, #customModal	localStorage.voiceTranscript, sessionStorage.voiceStudioTransfer
Files (files/files.js)	activeFolderId, foldersCache, currentPreviewFile, activeAnalyses (Map), toastMinimized	#folderList, #newFolderBtn, #uploadDropzone, #fileInput, #fileGrid, #fileSearch, #viewGrid, #viewList, #previewModal, #analysisToast, #modalLiveThinkingText	None (DB-backed via tRPC)
Image Studio (studio/studio.js)	currentJobId, activeEventSource, pollTimer, isRatioLocked, lockedRatio, galleryItems	#modelSelect, #upscaleSettingsGroup, #upscaleModelSelect, #samplerSelect, #widthInput, #heightInput, #lockRatioToggle, #generateBtn, #progressBarFill, #canvasSplit, #basePane, #finalPane, #galleryGrid, #imageModal	None (DB-backed via image.list)
FramesX (framesx/framesx.js)	currentJobId, activeEventSource, durationSec, SINGLE_SCENE_MAX=6, galleryVideos	#videoProviderSelect, #planningModelSelect, #durationSlider, #multiSceneNotice, #planningStreamText, #sceneCardsContainer, #resultVideo, #videoGalleryGrid	None (DB-backed via /api/video/list)
Voice Studio (voice/voice.js)	9 OOP classes: _playToken, _pending (Map), _nextSeq, currentStage, currentRgb, handsFree	#voiceCanvas, #stateText, #modeTtsBtn, #modeSttBtn, #modeStsBtn, #ttsInput, #sttRecordBtn, #stsMicBtn, #voiceChatStream, #settingsDrawer, #themeSelector	sessionStorage.voiceStudioTransfer
Persona Studio (personastudio.js)	voicesList, personasList, setupStatus, serverStatus, pollInterval	#serverStateContainer, #activeLogConsole, #hfToken, #cpuOffload, #personaForm, #personaName, #personaVoice, #personaPrompt, #personasGrid	None (personas.json + tRPC)
The Forge (forge/forge.js)	currentSessionId, activeEventSource, rawInThinkBlock, rawTagBuffer, tasksCache	#liquidField, #particles, #sessionList, #goalInput, #stackSelect, #modelSelect, #taskTree, #rawStreamBody, #liveLogBody, #followupInput, #pauseResumeBtn	None (DB-backed via forge.*)
ForgeX (forgex/forgex.js)	currentSessionId, activeEventSource, sessionsCache	#sessionList, #forgexGoal, #forgexModel, #forgexTerminal, #forgexFollowupInput, #forgexStopBtn	None (DB-backed via forgex.*)
Monitor (monitor/monitor.js)	uptimeSeconds, 4 polling intervals (1s/3s/5s/15s)	#cpuVal, #gpuVal, #ramVal, #vramVal, #cpuTemp, #gpuTemp, #netUp, #netDown, #processList, #serviceList, #storageList, #eventTerminal, #uptime	None (real-time /api/system/*)
Settings (settings/settings.js)	loadedSections (Set), wmCurrentWorkerId, wmLogEventSource, wmPollTimer	#settingsNav, #settingsSearch, #activeModelsList, #installedModelsList, #keepAliveToggle, #pythonStartBtn, #speechStartBtn, #clusterGrid, #workerModal, #wmLogPane	None (lazy-loaded tRPC tabs)
Model Manager (model-manager.js)	EventManager listeners, activeCategory, searchQuery, activeDownloads	#category-list, #overview-grid, #package-grid, #model-search, #download-dock-root	In-memory StateManager
Robotics (robotics/robotics.js)	2500ms sensor simulation interval, resize observer	#topologySvg, #n-core, #simTemp, #simHum, #simVoc	None (simulation alpha)

PART IV — FRONTEND-TO-BACKEND WIRE PROTOCOLS

SuperJSON tRPC Envelopes, NDJSON Streams, SSE & Multipart Uploads

Complete Contract Reference Linking Frontend Call Sites to Backend Handlers

1. Native Fetch over tRPC v11 SuperJSON Protocol

Because the production frontend is a Vanilla ES6+ MPA without the React tRPC client bundle, every module invokes tRPC procedures directly over HTTP using tRPC v11's **SuperJSON wire envelope** (configured in **d:/AI/Chatbot/app/api/middleware.ts #L1-L11** via *initTRPC.context<Context>().create({ transformer: superjson })*):

```
// 1. tRPC Query (HTTP GET) — e.g. settings.js #L22-L31 & forge.js #L287-L298
async function trpcQuery(proc, input) {
  const qs = input !== undefined
    ? '?input=' + encodeURIComponent(JSON.stringify({ json: input }))
    : '';
  const res = await fetch('/api/trpc/' + proc + qs);
  const data = await res.json();
  return data?.result?.data?.json;
}

// 2. tRPC Mutation (HTTP POST) — e.g. settings.js #L33-L45 & forge.js #L300-L309
async function trpcMutate(proc, input) {
  const res = await fetch('/api/trpc/' + proc, {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ json: input ?? {} })
  });
  const data = await res.json();
  return data?.result?.data?.json;
}
```

2. Complete REST, NDJSON Streaming & SSE Endpoint Matrix

Endpoint & Method	Protocol	Frontend Caller	Backend Handler	Behavior & Stream Events
`POST /api/chat/stream`	NDJSON Stream	`chat.js #L294, voice.js #L139`	`boot.ts #L471-L1089`	Standard chat + 5-stage tool router + RAG + memory + optional Kokoro sentence TTS. Emits {thinking}, {status}, {response}, {audioSegment, seq}, {error}.
`POST /api/chat/agent-stream`	NDJSON Stream	`chat.js #L483`	`boot.ts #L1091-L1416`	25-round Extended Thinking agent loop. Emits {round}, {thinking}, {tool_call}, {tool_result}, {summary_chunk}, {summary_done}, {response}, {error}.
`GET /api/chat/sources`	REST JSON	`chat.js #L432`	`boot.ts #L430+`	Returns RAG source citations for a conversationId.
`POST /api/chat-files/upload`	Multipart FormData	`chat.js #L225`	`chatUpload.ts #L1-L203`	Uploads chat attachments, extracts text, chunks (1000 chars), embeds via nomic-embed-text, and stores in messageAttachments.
`POST /api/files/upload`	Multipart FormData	`files.js #L314`	`upload.ts #L1-L175`	Saves file to uploads/, inserts files row, triggers runKnowledgeWorkflow(id) for extraction + chunking + embedding.
`GET /api/files/:id/analyze-stream`	SSE Stream	`files.js #L604`	`analyzeStream.ts #L1-L199`	Streams 5-node LangGraph analysis (progress, thinking, done, error) from GPU worker or Python runtime (:8002) and upserts documentAnalysis.
`GET /api/files/:id/analysis`	REST JSON	`files.js #L803`	`analyzeStream.ts #L179-L197`	Fetches persisted LangGraph analysis report from MySQL.
`POST /api/image/generate & GET /api/image/stream/:jobId`	REST + SSE	`studio.js #L417, #L454`	`boot.ts / image.ts`	Starts ComfyUI single-pass or 2-pass Hires job and streams queued, comfy_node, comfy_progress, done, error.
`POST /api/video/generate & GET /api/video/stream/:jobId`	REST + SSE	`framesx.js #L397, #L506`	`boot.ts / framesx.ts`	Runs LLM scene planning + 3-stage LTX-Video / Wan2.1 ComfyUI rendering + FFmpeg concat. Emits 14 named SSE events.
`/api/speech/* & /api/voice/*`	REST + Stream	`voice.js #L56-L137`	`boot.ts #L1425-L2189`	TTS synthesis, STT transcription, voice cloning profiles, and sentence-pipelined STS voice chat.
`GET /api/forge/:id/stream`	EventSource SSE	`forge.js #L518`	`forge/routes.ts #L38-L166`	Streams real-time model tokens (token) + 1s DB poll (iteration, status, raw_activity, error).
`GET /api/forgex/:id/stream`	EventSource SSE	`forgex.js #L244`	`forgex/routes.ts #L20-L96`	Streams headless Claude Code CLI stdout/stderr/system output and status transitions.
`GET /api/cluster/:id/logs/stream`	EventSource SSE	`settings.js #L528`	`boot.ts #L406-L469`	Proxies live stdout/stderr ring buffer stream from remote GPU cluster worker (:3001/logs/stream).

PART V — BACKEND CORE, SERVER BOOT & TRPC ROUTERS

Hono Server Initialization, Background Collectors & All 17 tRPC Routers

Deep Walkthrough of api/boot.ts, api/router.ts, api/queries/connection.ts & api/routers/*

1. Server Boot Sequence & Background Daemons (api/boot.ts #L60-L275, #L2536-L2629)

When **d:/AI/Chatbot/app/api/boot.ts** starts (either under *@hono/vite-dev-server* in development or *node dist/boot.js* in production), it executes a deterministic 7-step initialization sequence:

1. **Environment & Hono Middleware (#L156-L185)**: Loads *.env* via *api/lib/env.ts*, creates the root *Hono* instance, mounts request logging middleware (#L162-L169), and registers a global *app.onError* handler (#L172-L185).
2. **Distributed Cluster Worker Poller (startWorkerPoller() #L2536)**: Starts a 5,000ms interval polling *GET {workerUrl}/capabilities* across *WORKER_1_URL..WORKER_10_URL* and *WORKER_DESKTOP_URL* (**api/services/workerPoller.ts #L131-L170**).
3. **Speech Management Subsystem (initializeSpeechSystem() #L2547-L2559)**: Lazily loads *SpeechManager* (**api/speech/lazy-speech-manager.ts #L20-L26**), registering all 7 TTS/STT providers, voice cloning profiles, and runtime health checks.
4. **Auto-DDL Table Creation (ensureDatabaseTables() #L60-L132)**: Executes idempotent *CREATE TABLE IF NOT EXISTS* statements for *document_analysis*, *forge_sessions*, *forge_task_nodes*, *forge_iterations*, *forgex_sessions*, and *forgex_output*.
5. **Database Seeding & Vector Backfill (initDatabase() #L134-L154)**: Calls *seedDatabase()* (**db/seed.ts #L12-L490**) if the *models* table is empty, then launches *backfillEmbeddings()* (**api/services/backfillEmbeddings.ts #L1-L67**) to generate 768-dim *nomic-embed-text* vectors for any unindexed rows in *fileContents* or *knowledgeMemories*.
6. **Hardware Telemetry Collector (startTelemetryCollector() #L262-L265)**: Starts a 15,000ms daemon (**api/services/telemetry.ts #L1-L105**) querying *nvidia-smi* and *systeminformation* to insert hardware snapshots into *systemMetrics*.
7. **ESP32 Companion Hardware Server (startCompanionServer() #L270-L274)**: Binds a dedicated HTTP + *WebSocketServer* on *0.0.0.0:3005* (**api/services/companion.ts #L601-L820**).

2. Complete Inventory of All 17 tRPC Sub-Routers (api/router.ts #L1-L47)

Router Key	Source File	Exposed Procedures & Technical Behavior
`model`	`api/routers/model.ts (#L1-L166)`	list (merges DB models + live Ollama /api/tags + /api/ps VRAM state + OpenRouter models), getById, load (POST /api/generate with keep_alive), unload (keep_alive: 0), updateConfig, setActive, pull, delete.
`conversation`	`api/routers/conversation.ts (#L1-L109)`	list (ordered by isPinned DESC, updatedAt DESC), getById (joins messages + messageToolCalls grouped by step/round), create, update, delete, togglePin, clearAll.
`message`	`api/routers/message.ts (#L1-L211)`	listByConversation, create (extracts fenced codeBlocks metadata, updates conversation token/message counters), createWithToolCalls (persists multi-round thinking + messageToolCalls rows), delete.
`chat`	`api/routers/chat.ts (#L1-L55)`	send (non-streaming chat mutation with RAG context injection).
`file`	`api/routers/file.ts (#L1-L276)`	list, getStats, create, delete (removes disk file + DB cascade), toggleStar, search (LIKE search over files.name + fileContents.extractedText), createFolder, listFolders, deleteFolder, moveToFolder.
`fileContent`	`api/routers/fileContent.ts (#L1-L33)`	getByFileId (returns extractedText, ocrText, visionAnalysis, tablesJson, metadata for the Files modal).
`image`	`api/routers/image.ts (#L1-L340)`	list, generate (validates provider, creates job in generationState, launches runGeneration), enhancePrompt (via Ollama LLM), getJobStatus, getActiveJobs, delete.
`video`	`api/routers/video.ts (#L1-L110)`	list, getById, delete (removes video file from public/uploads/videos/ and DB row).
`providers`	`api/routers/providers.ts (#L1-L69)`	list (filters image/video providers from services/providers.ts), get, comfyOptions (queries live ComfyUI /object_info for KSampler samplers/schedulers and UpscaleModelLoader models).
`speech`	`api/speech/index.ts (#L1-L49)`	Nested routers: tts, stt, providers, models, voices (cluster-wide voice aggregation), profiles (voice cloning), downloads, devices, benchmark, voiceChat, runtime.
`personaplex`	`api/routers/personaplex.ts (#L1-L400)`	voices (18 Moshi embeddings), personas.{list,create,update,delete} (personas.json), setup.{status,start} (6-step git clone + venv + CUDA torch install), server.{status,start,stop} (port 8998 moshi.server).
`forge`	`api/forge/router.ts (#L1-L180)`	listModels, listStacks (5 stack profiles), create (allocates workspace & port, runs DAG planning, starts loop), pause, resume, followUp, getStatus, getTaskTree, getTaskLog, list.
`forgex`	`api/forgex/router.ts (#L1-L101)`	listModels (filters local tool-capable Ollama models), create (spawns claude -p), sendInput (spawns claude -p --resume), stop (kills child PID), getStatus, list.
`system`	`api/routers/system.ts (#L1-L138)`	getCurrent, getHistory, getServiceStatus (checks MySQL, Ollama :11434, ComfyUI :8188, Python :8002, Speech :9000), getProcesses (ps-list + pidusage top 15), getStorage (si.fsSize).
`runtime`	`api/routers/runtime.ts (#L1-L26)`	python.status, python.start, python.stop (controls local FastAPI Python runtime on 127.0.0.1:8002 via services/runtime/manager.ts).
`cluster`	`api/routers/cluster.ts (#L1-L69)`	workers (lists registered GPU nodes), worker (single worker details), metricHistory (180-point ring buffer from clusterMetricsHistory.ts), logs (fetches worker stdout/stderr snapshot).
`knowledge / research / workflow / activity`	`api/routers/{knowledge,research,workflow,activity}.ts`	Long-term memory CRUD (knowledgeMemories), research collections (researchCollections/Documents), workflow graph CRUD & execution (workflows/workflowRuns), and audit logs (activityLogs).

PART VI — CHAT, HYBRID RAG, MEMORY & AGENT LOOP

Context Assembly, 5-Stage Tool Router, FenceWatcher & PhaseTrackerService

Deep Technical Analysis of api/services/{agentLoop,toolRouter,toolPattern,phaseTracker,rag,semanticSearch}.ts

1. Multi-Layer Prompt Context Assembly Pipeline

Every chat request in **POST /api/chat/stream** (#L471-L1089) and **POST /api/chat/agent-stream** (#L1091-L1416) constructs a rich, multi-layered system/context prompt before invoking Ollama or OpenRouter:

- **Layer 1 — Device-Synced Temporal Context (api/lib/temporalContext.ts #L60-L109)**: *buildTemporalContext({ timezone, conversationId })* formats the current date, day-of-week, and local time using *Intl.DateTimeFormat* in the client's exact IANA timezone, queries the last two message timestamps in MySQL, formats the elapsed gap via *formatGap(ms)* (#L18-L45, e.g., *'3 hours 14 minutes ago'*), and injects *[TEMPORAL CONTEXT]*.
- **Layer 2 — Semantic Long-Term Memory Retrieval (api/services/memorySemanticSearch.ts #L12-L146)**: *searchRelevantMemories(query, limit = 5)* embeds the prompt via *nomic-embed-text* (*http://localhost:11434/api/embeddings*), scores all rows in *knowledgeMemories* using **cosine similarity + keyword overlap boost + importance weighting** (filtering *score > 0.45*, or returning top memories if total count <= 10), updates *lastAccessed = new Date()*, and formats *[USER MEMORY CONTEXT]*.
- **Layer 3 — Token-Budgeted History & Rolling Summarization (api/services/contextWindow.ts #L1-L209 & conversationSummary.ts #L1-L105)**: *buildContextWindow()* enforces *MAX_CONTEXT_TOKENS = 150,000* (estimated at *CHARS_PER_TOKEN = 3.5*) and *MAX_CONTEXT_MESSAGES = 50*. Whenever a conversation's message count hits a multiple of 20 (*count % 20 === 0*), *updateConversationSummary()* (#L16-L95) asynchronously summarizes the latest 25 messages via local Ollama (*gemma3:4b*) and saves the rolling summary into *conversations.summary*.
- **Layer 4 — Hybrid Semantic + Keyword RAG (api/services/rag.ts #L1-L214 & semanticSearch.ts #L12-L266)**:
  - *searchKnowledge(query, selectedFileIds)* embeds the query with *nomic-embed-text* and scores every chunk in *fileContents* using the hybrid formula: **finalScore = (0.8 * cosineSimilarity) + (0.2 * keywordMatchRatio)** (#L120-L155).
  - Applies a **dynamic relative cutoff**: *threshold = Math.max(0.35, bestScore - 0.15)* (#L162), capping at **max 2 chunks per file** and top 6 chunks overall so a single large PDF cannot monopolize the context window.
  - Enriches matched files with structured outputs from *fileContents.visionAnalysis*, *ocrText*, and *tablesJson*.
- **Layer 5 — Conversation Attachment RAG (api/services/chatAttachmentSearch.ts #L14-L130)**: *searchChatAttachments(conversationId, query)* ranks embedded chunks from *messageAttachments* for files uploaded directly inside the current chat thread (threshold > 0.35, top 5 chunks).

2. The 5-Stage Tool Router Cascade (api/services/toolRouter.ts #L109-L701)

To avoid slow LLM tool-calling overhead on conversational prompts while maintaining high recall on tool-requiring prompts, *shouldUseTools(message, hasAttachments)* (#L586-L701) runs a 5-stage cascade:

Figure 6.1 — 5-Stage Tool Router Cascade (api/services/toolRouter.ts & toolPattern.ts)



- **Stage 0 — Cheap Pre-Check (cheapPreCheck #L109-L119)**: Immediately returns *{ useTools: false }* if the trimmed prompt is < 4 characters or matches *GREETING_RE* (e.g., 'hi', 'hello', 'thanks', 'ok') and has no attachments.
- **Stage 1 — Fast Binary LLM Gate (stage1YesNoGate #L150-L227)**: Prompts *ROUTER_MODEL = 'lfm2.5-thinking:1.2b'* (*temperature: 0*, *num_predict: 120*, *8,000ms* AbortController timeout) to strip *<think>* tags and answer strictly *yes* or *no*.

Fails safe to *true* if the router model times out.

- **Stage 2 — Deterministic Regex Pattern Scorer (matchPatterns in toolPattern.ts #L434-L467)**: Evaluates weighted regex patterns across all 18 tools (*TOOL_PATTERNS* #L47-L348). If a single tool matches with high confidence and does NOT belong to *CONFUSABLE_CLUSTERS* (#L356-L384: e.g., *['knowledge_search', 'file_search', 'internet_search', 'research_query']* or *['local_vision_ocr', 'layout_analyzer', 'local_vision_analyzer']*) and has *needsDisambiguation: false*, it synthesizes arguments immediately via *buildDefaultArguments()* (#L495-L509) without another LLM call!
- **Stage 3 — Schema Fill-In & Multi-Format Fallback (stage3FillIn #L347-L529)**: Invokes *lfm2.5-thinking:1.2b* via Ollama */api/chat* with native *tools* JSON schemas. If the model emits text instead of native *message.tool_calls*, the parser falls back to extracting *<tools>...</tools>* XML blocks (*extractToolsTagPayload* #L243-L280) or bare JSON tool objects (*extractBareJsonToolCalls* #L286-L345).

3. Extended Thinking 25-Round Agent Loop (api/services/agentLoop.ts #L1-L1889)

When Extended Thinking is enabled in Chatspace (*POST /api/chat/agent-stream -> runAgentLoop* #L1346-L1889), the backend runs an autonomous multi-round reasoning, tool-execution, and phase-verification state machine:

- **Capability Verification (modelSupportsThinking #L81-L123)**: Queries Ollama *POST /api/show* for *capabilities.includes('thinking')* or matches known reasoning families (*qwen3*, *deepseek-r1*, *qwq*, *lfm2.5-thinking*, *cogito*).
- **Real-Time FenceWatcher Interceptor (FenceWatcher class #L660-L783)**: As tokens stream from the LLM in each round, *FenceWatcher.push(chunk)* buffers potential backtick sequences. The moment it detects an opening ```` ```tool_call ```` fence, it withholds those characters from the user-facing *response* stream, buffers the JSON body, and **immediately aborts the LLM stream** as soon as the closing ```` ``` ```` arrives so the backend can execute the tool without waiting for trailing hallucinated prose.
- **3-Layer Unfenced Tool Recovery Cascade (#L1470-L1545)**: If a model finishes a round without emitting a ```` ```tool_call ```` fence, the loop inspects the output through 3 fallback extractors:
  1. *tryExtractInstructionFromProse()* (#L306-L312): Matches inline *tool_call { ... }* JSON blocks.
  2. *tryExtractInstructionFromPlainFences()* (#L354-L406): Parses standard ```` ```json ```` or ```` ``` ```` blocks containing tool names.
  3. *resolveInstructionViaSchema()* (#L160-L214) -> *resolveAgentToolCall()* (**toolRouter.ts #L727-L827**): Sends natural-language tool intent to *JSON_EXTRACTOR_MODEL = 'lfm2.5:8b'* with a strict JSON schema to produce a valid *{ tool, arguments }* object.
- **Duplicate Call Guard (executedCalls Set #L1411-L1418)**: Hashes every executed *`${toolName}::${JSON.stringify(args)}`*. If the model attempts the exact same tool call again, the loop blocks re-execution and injects a system directive forcing the model to advance to the next step.
- **PhaseTrackerService & Premature Final-Answer Gate (api/services/phaseTracker.ts #L178-L536)**:
  – After Round 1 thinking, *PhaseTrackerService.extractPlan()* (#L210-L310) uses *lfm2.5-thinking:1.2b* to extract a structured array of execution phases (*id*, *title*, *requiredTools*, *status*).
  – On subsequent rounds, it injects *[EXECUTION PLAN STATUS]* into the prompt and marks phases *completed* as matching tools succeed.
  – **Completeness Gate (agentLoop.ts #L1557-L1585)**: If the model attempts to stop and output a final answer while planned phases are still *pending* or *in_progress*, the loop rejects the final answer (up to *MAX_FINAL_ANSWER_REJECTIONS = 2*) and forces another round specifying the exact unfinished phases.
- **Executive Post-Loop Synthesis (synthesizeExecutionSummary #L1170-L1340)**: When an agent run executes >= 3 tool calls, the loop runs a dedicated synthesis pass producing structured *## Executive Summary*, *### Key Findings*, and *### Detailed Breakdown* sections.

4. Complete Specification of All 18 Agent Tools (api/services/tools.ts & toolExecutor.ts)

Tool Name	Implementation Target	Arguments Schema	Technical Execution Details
knowledge_search	semanticSearch.ts #L12	{ query, limit? }	Hybrid 0.8*semantic + 0.2*keyword search over fileContents.
file_search	routers/file.ts	{ query }	Searches files by filename and extractedText in MySQL.
internet_search	tavily.ts #L1-L84	{ query, max_results? }	Queries https://api.tavily.com/search with search_depth:'advanced'.
research_query	research.ts #L1-L81	{ query }	Runs Tavily search + scrapes full Markdown of top 3 URLs via Firecrawl.
url_reader	firecrawl.ts #L1-L91	{ url }	Scrapes clean Markdown from any URL via @mendable/firecrawl-js.
local_vision_ocr	Python :8002 / GPU Worker	{ image_path }	Runs Surya OCR (SuryaProvider) to extract text lines & bounding boxes.
layout_analyzer	Python :8002 / GPU Worker	{ image_path }	Runs Surya Layout detection (tables, figures, headers, paragraphs).
marker_pdf_pipeline	Python :8002 / GPU Worker	{ pdf_path }	Converts complex PDFs to Markdown + extracts structured tables via Marker.
local_vision_analyzer	Python :8002 / GPU Worker	{ image_path, prompt }	Runs multimodal visual Q&A via Qwen3-VL or MiniCPM-V on Ollama.
document_analysis	Python :8002 / GPU Worker	{ text, file_id? }	Executes the 5-node LangGraph document intelligence pipeline.
memory_store	memory.ts #L1-L312	{ content, category?, importance? }	Embeds via nomic-embed-text and inserts into knowledgeMemories.
memory_search	memorySemanticSearch.ts	{ query, limit? }	Semantic vector search over long-term user memories.
memory_update	memory.ts	{ id, content, importance? }	Updates an existing memory row and recomputes its embedding.
memory_delete	memory.ts	{ id }	Deletes a memory row from knowledgeMemories.
scientific_calculator	toolExecutor.ts #L165	{ expression }	Evaluates math expressions safely in strict JS sandbox with Math.* constants.
system_monitor	telemetry.ts #L1-L105	{}	Returns live GPU, VRAM, CPU, RAM, disk, and temperature metrics.
model_manager / ollama_control	modelManager.ts / ollamaControl.ts	{ action, model? }	Lists, loads, or unloads models in local Ollama VRAM.
sql_query	sqlQuery.ts #L1-L31	{ query }	Executes read-only SELECT queries against the workstation MySQL database.

PART VII — AI RUNTIMES, MEDIA, FORGE AGENTS & GPU CLUSTER

Python (:8002), Speech (:9000), ComfyUI (:8188), Forge/ForgeX, Worker (:3001) & ESP32 (:3005)

Complete Engineering Breakdown of Microservices, LangGraph, Multi-Scene Video & IoT Hardware

1. Python Tool Runtime (127.0.0.1:8002) & 5-Node LangGraph Pipeline

Files: *d:/AI/Chatbot/app/python-runtime/app/main.py* (#L1-L79), *runtime.py* (#L1-L155), *tools/document_analysis/graph.py* (#L1-L278), and Node bridge *d:/AI/Chatbot/app/api/services/analyzeStream.ts* (#L1-L199).

- **On-Demand Spawner & Cluster Offloading (api/services/runtime/manager.ts #L1-L187)**: When a Python tool or document analysis is requested, Node first checks *selectWorker(toolName)* (**api/services/cluster.ts #L184-L214**) to offload heavy GPU work to an online cluster worker. If no worker is online, *RuntimeManager.ensureRunning('python')* spawns *uvicorn app.main:app --host 127.0.0.1 --port 8002* and polls *GET /health* until ready.
- **5-Node LangGraph StateGraph (python-runtime/app/tools/document_analysis/graph.py #L114-L278)**: Compiled with *StateGraph(DocumentAnalysisState)* using local Ollama *MODEL = 'lfm2.5:8b'* (*format: 'json'*, *stream: True*):
  1. *ingestion_node* (#L114-L121): Validates raw text and initializes state.
  2. *summarizer_node* (#L124-L135): Generates executive summary via *SUMMARY_PROMPT*.
  3. *extraction_node* (#L138-L149): Extracts structured *entities* (*people*, *organizations*, *locations*, *dates*, *technologies*), *keywords*, and *topics*.
  4. *metadata_node* (#L152-L166): Classifies *language*, *document_type*, and *confidence* score (0.0–1.0).
  5. *synthesizer_node* (#L169-L192): Merges outputs into the final report dictionary.
- **Asyncio Queue Token Streaming & MySQL Upsert**: While each node calls Ollama, an *_on_token* callback pushes live tokens into an *asyncio.Queue*, yielding SSE *event: thinking* and *event: progress* frames to *analyzeStream.ts*, which forwards them to the browser and upserts the completed JSON into *documentAnalysis* (#L140-L170).

2. ComfyUI Image Studio & FramesX Multi-Scene Video Pipelines (:8188)

- **2-Pass Hires Image Pipeline (api/services/comfyWorkflowFull.ts #L1-L265 & backgroundGeneration.ts #L1-L145)**:
  – Supports 4 providers in **api/services/providers.ts** (#L20-L92): *flux1-schnell*, *flux2-klein*, *flux2-klein-full*, and *flux1-krea-dev-gguf*.
  – For *flux2-klein-full*, *convertLiteGraphToApi()* (#L70-L152) converts ComfyUI's UI-format LiteGraph JSON into API execution format, and *buildFluxKleinFullWorkflow()* (#L198-L265) patches Pass 1 generation + ESRGAN pixel/latent upscaling (*4x-UltraSharp.pth*) + Pass 2 Hires refinement sampler/denoise parameters, saving both Pass 1 (*baseImageUrl*) and Pass 2 images.
- **FramesX 3-Stage Hierarchical Video Director (api/services/framesx.ts #L1-L1036)**:
  1. **LLM Scene Expansion (planScenes #L141-L234)**: Prompts *qwen2.5:7b-instruct* on Ollama to expand a user prompt into *N* sequential cinematic scene descriptions, streaming tokens live via *planning_token* SSE events.
  2. **3-Stage LTX-Video 2B Execution (buildLTX3StageSceneWorkflow #L478-L575 & renderSceneLTX3Stage #L581-L708)**: Renders each scene through Stage 1 (Half-Res Draft 384x256) -> Stage 2 (Latent Upscale 768x512 + *LTXVAdainLatent*) -> Stage 3 (Temporal Smoothing + *SaveWEBM*).
  3. **FFmpeg Last-Frame Extraction & Conditioning (#L714-L746)**: Runs *ffmpeg -sseof -0.1 -i scene.webm -frames:v 1 last_frame.png* (*extractLastFrame*), uploads the PNG to ComfyUI *POST /upload/image*, and injects it into the next scene's *LTXVImgToVideo* node for continuous visual transitions across scenes.
  4. **FFmpeg Concatenation (mergeScenes #L753-L821)**: Transcodes animated WebP/WebM clips if needed and concatenates all scene files via *ffmpeg -f concat -safe 0 -i list.txt -c copy*.

3. Speech Subsystem (:9000), Cluster Voice Aggregation & PersonaPlex (:8998)

- **7 Providers & Distributed Voice Aggregation (api/speech/manager/providerManager.ts #L272-L355)**: Manages *kokoro*, *whisper*, *piper*, *xtts*, *fishspeech* (:6101), *dia*, and *chatterbox* (:6102). *listVoices()* merges local voices with remote voices queried from all online cluster workers (*POST {workerUrl}/speech*), deduplicating by *providerId::modelId::voiceId*.
- **2-Worker Sentence-Pipelined Voice Chat (boot.ts #L1653-L2189 & api/speech/audio/wav.ts #L78-L91)**: In *POST /api/voice/chat-stream*, LLM tokens are split on sentence boundaries (*.!?*) and dispatched into a 2-worker TTS pool (*TTS_CONCURRENCY = 2*) so sentence *N+1* synthesizes while sentence *N* plays. At turn completion, *WavUtil.concat()* strips the 44-byte RIFF/WAV headers from subsequent chunks and writes a single combined WAV file.
- **NVIDIA PersonaPlex Full-Duplex Server (api/routers/personaplex.ts #L1-L400)**: Automates 6-step setup (git clone, venv, PyTorch CUDA install) and spawns *python -m moshi.server --ssl <tempDir>* (with optional *--cpu-offload*) on port *8998*.

4. Autonomous Coding Agents: The Forge & ForgeX

- **The Forge (api/forge/orchestrator.ts #L1-L229, agentLoop.ts #L1-L581, execWorker.ts #L1-L238)**:
  - Isolates each session in *D:\Forge\workspaces\{sessionId}* with an OS-allocated free TCP port (*findFreePort* in *db.ts #L104-L120*).
  - Plans a DAG of tasks (*forgeTaskNodes*) and runs a **THINK -> ACT -> EVALUATE** loop per task. *contracts.ts* (#L31-L64) propagates exported file contracts down the dependency tree.
  - **Security Sandboxing (execWorker.ts #L40-L90)**: *resolveSafePath()* blocks any path resolving outside the workspace root, and *validateCommand()* blocks destructive shell commands (*rm -rf*, *del /s*, *format*, *shutdown*, *reg*). Shell commands (*run_command*) are structurally restricted to the final mandatory stack *isIntegrationCheck: true* task (#L296-L319).
- **ForgeX — Headless Claude Code CLI Bridge (api/forgex/processManager.ts #L1-L611)**:
  - Writes *D:\ForgeX\workspaces\{sessionId}\.claude\settings.json* (#L182-L225) with *ANTHROPIC_BASE_URL: 'http://localhost:11434'*, *ANTHROPIC_AUTH_TOKEN: 'ollama'*, and *ANTHROPIC_MODEL: modelId*.
  - Spawns *claude -p "<goal>" --output-format stream-json --verbose --permission-mode acceptEdits* with *stdio: ['ignore', 'pipe', 'pipe']* (closing stdin immediately so Claude Code does not hang waiting for EOF). Captures *session_id* from the *init* JSON event and passes *--resume <claudeSessionId>* on all subsequent follow-up turns (#L566-L589).

5. Distributed GPU Cluster (:3001/:3443) & ESP32 Companion Server (:3005)

- **Cluster Poller & Base64 File Marshaling (workerPoller.ts #L1-L170, workerClient.ts #L58-L227)**: Master polls worker */capabilities* every 5s, stores a 180-point telemetry ring buffer (*clusterMetricsHistory.ts*), and routes jobs to the lowest-GPU worker via *selectWorker()*. Before sending vision/PDF jobs over the network, *injectFileData()* (#L58-L99) reads local disk paths (*image_path*, *pdf_path*) and injects base64 *image_data* / *pdf_data* so remote workers require zero shared storage.
- **Worker Node & Cloudflare Tunnel Gateway (worker/src/server.ts #L1-L413 & exposureService.ts #L1-L663)**: Port 3001 worker supervises local Python (:8002) and Speech (:9000) runtimes with 3-second crash auto-restart timers. Port 3443 *exposureService.ts* manages *cloudflared* tunnels (*quick*, *named*, *dashboard*) with a 3-state machine (*AUTO*, *FORCE_ON*, *FORCE_OFF*), Bearer API key auth, 100 req/min rate limiting, and MySQL audit tables.
- **ESP32 Companion Server (api/services/companion.ts #L1-L820)**: Listens on *0.0.0.0:3005*. Handles REST + WebSocket frames (*hello*, *ping*, *chat*, *voice_start*, binary PCM/WAV chunks, *voice_end*), runs Whisper STT -> *gemma3:4b* -> Kokoro *af_sarah* TTS, streams audio back in 8KB *voice_audio_chunk* frames, and broadcasts 10-minute cached OpenWeatherMap snapshots.

6. Deep-Dive Reference: Forge Stack Profiles, ComfyUI Graphs & Companion WS Protocol

Stack Profile ID	Source Definition	Conventions & Mandatory Integration Check Behavior
`node-express`	`api/forge/stacks/nodeExpress.ts`	Node.js + Express + TypeScript/ESM REST API profile. Integration check verifies npm install, tsc/syntax check, server boot on allocatedPort, and HTTP health check.
`java-spring-boot`	`api/forge/stacks/javaSpringBoot.ts`	Java 17+ / Maven or Gradle Spring Boot profile. Integration check runs mvn compile / test and verifies application context startup.
`mern-stack`	`api/forge/stacks/mernStack.ts`	Full-stack MongoDB/Express/React/Node profile with coordinated backend API + frontend client build verification.
`python-fastapi`	`api/forge/stacks/pythonFastapi.ts`	Python FastAPI + Uvicorn + Pydantic profile. Integration check validates syntax (py_compile), requirements, and Uvicorn boot on allocatedPort.
`general`	`api/forge/stacks/general.ts`	Language-agnostic profile supporting arbitrary scripts, CLI tools, and static web apps with custom stack instructions.

WebSocket Frame / Direction	Payload Shape	Backend Processing Pipeline (api/services/companion.ts)
`Client -> Server: hello`	`{ type: 'hello', deviceId, name, deviceType, capabilities, fwVersion }`	Registers device in companion/registry.ts (#L1-L290) andcompanionDevices table; replies with welcome + ai_status + weather_update.
`Client -> Server: ping`	`{ type: 'ping', uptimeMs?, rssi?, freeHeap? }`	Updates device lastSeen & telemetry; server replies with { type: 'pong', serverTime }.
`Client -> Server: chat`	`{ type: 'chat', text, sessionId? }`	Streams LLM response via gemma3:4b (COMPANION_LLM_MODEL) emitting chat_token and chat_done frames.
`Client -> Server: voice_start`	`{ type: 'voice_start', format: 'pcm16'|'wav', sampleRate: 16000 }`	Initializes per-connection audio chunks buffer in companion.ts (#L650-L710).
`Client -> Server: [Binary Frame]`	`Raw PCM16 / WAV binary WebSocket frame`	Appended to active voice buffer for the connected ESP32 or browser client.
`Client -> Server: voice_end`	`{ type: 'voice_end' }`	Wraps PCM in 44-byte WAV header if needed, runs Whisper STT -> emits voice_transcript -> runs gemma3:4b -> emits voice_response -> synthesizes Kokoro af_sarah WAV -> streams 8KB voice_audio_chunk frames -> emits voice_done.

PART VIII — SOURCE-VERIFIED SCHEMA & CODEBASE INDEX

Complete 34-Table Column Dictionary & Full-Stack File/Symbol Inventory

Automatically Introspected from d:/AI/Chatbot/app/db/schema.ts, frontend/*, api/*, python-runtime/* & worker/*

1. Complete 34-Table MySQL / Drizzle ORM Schema Dictionary (db/schema.ts)

Every table and column below is parsed directly from **d:/AI/Chatbot/app/db/schema.ts** (#L1-L872), including exact Drizzle column types, SQL constraints, defaults, and line numbers:

1.1 Table users (Drizzle Export: *users* — db/schema.ts #L16-L23)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L17
email	varchar("email", { length: 255 })	.notNull().unique()	L18
name	varchar("name", { length: 255 })	nullable	L19
avatarUrl	varchar("avatar_url", { length: 500 })	nullable	L20
role	mysqlEnum("role", ["admin", "user"])	.default("admin")	L21
createdAt	timestamp("created_at")	.defaultNow()	L22

1.2 Table folders (Drizzle Export: *folders* — db/schema.ts #L25-L30)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L26
name	varchar("name", { length: 255 })	.notNull()	L27
userId	bigint("user_id", { mode: "number", unsigned: true })	.references(() => users.id)	L28
createdAt	timestamp("created_at")	.defaultNow()	L29

1.3 Table conversations (Drizzle Export: *conversations* — db/schema.ts #L32-L43)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L33
title	varchar("title", { length: 255 })	nullable	L34
summary	text("summary")	nullable	L35
summarizedMessageCount	int("summarized_message_count")	.notNull().default(0)	L36
modelId	varchar("model_id", { length: 100 })	.notNull()	L37
userId	bigint("user_id", { mode: "number", unsigned: true })	.references(() => users.id)	L38
folderId	bigint("folder_id", { mode: "number", unsigned: true })	.references(() => folders.id)	L39
isPinned	boolean("is_pinned")	.default(false)	L40
createdAt	timestamp("created_at")	.defaultNow()	L41
updatedAt	timestamp("updated_at")	.defaultNow()	L42

1.4 Table messages (Drizzle Export: *messages* — db/schema.ts #L45-L78)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L46
conversationId	bigint("conversation_id", { mode: "number", unsigned: true })	.references(() => conversations.id)	L47
role	mysqlEnum("role", ["user", "assistant", "system"])	.notNull()	L49
content	text("content")	.notNull()	L50
thinking	longtext("thinking")	nullable	L58
isExtended	boolean("is_extended")	.default(false)	L65
executionSummary	longtext("execution_summary")	nullable	L74
tokensUsed	int("tokens_used")	.default(0)	L75
responseTime	int("response_time")	nullable	L76
createdAt	timestamp("created_at")	.defaultNow()	L77

1.5 Table message_tool_calls (Drizzle Export: *messageToolCalls* — db/schema.ts #L91-L122)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L92
messageId	bigint("message_id", { mode: "number", unsigned: true })	.notNull() .references(() => messages.id)	L93
round	int("round")	.notNull()	L100
instruction	text("instruction")	nullable	L105
tool	varchar("tool", { length: 100 })	.notNull()	L106
arguments	json("arguments")	nullable	L107
result	longtext("result")	nullable	L112
success	boolean("success")	.notNull().default(true)	L113
error	text("error")	nullable	L114
thinking	longtext("thinking")	nullable	L120
createdAt	timestamp("created_at")	.defaultNow()	L121

1.6 Table ai_models (Drizzle Export: *aiModels* — db/schema.ts #L124-L138)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L125
name	varchar("name", { length: 255 })	.notNull()	L126
displayName	varchar("display_name", { length: 255 })	.notNull()	L127
size	varchar("size", { length: 50 })	nullable	L128
quantization	varchar("quantization", { length: 50 })	nullable	L129
contextLength	int("context_length")	nullable	L130
status	mysqlEnum("status", ["active", "idle", "loading", "error"])	.default("idle")	L131
memoryUsage	bigint("memory_usage", { mode: "number" })	.default(0)	L132
vramUsage	bigint("vram_usage", { mode: "number" })	.default(0)	L133
tokenSpeed	float("token_speed")	.default(0)	L134
lastUsed	timestamp("last_used")	nullable	L135
isInstalled	boolean("is_installed")	.default(false)	L136
createdAt	timestamp("created_at")	.defaultNow()	L137

1.7 Table system_snapshots (Drizzle Export: *systemSnapshots* — db/schema.ts #L140-L156)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L141
cpuUsage	float("cpu_usage")	.notNull()	L142
cpuTemp	float("cpu_temp")	nullable	L143
ramUsage	float("ram_usage")	.notNull()	L144
ramTotal	bigint("ram_total", { mode: "number" })	nullable	L145
gpuUsage	float("gpu_usage")	nullable	L146
gpuTemp	float("gpu_temp")	nullable	L147
vramUsage	float("vram_usage")	nullable	L148
vramTotal	bigint("vram_total", { mode: "number" })	nullable	L149
storageUsage	float("storage_usage")	nullable	L150
networkRx	bigint("network_rx", { mode: "number" })	nullable	L151
networkTx	bigint("network_tx", { mode: "number" })	nullable	L152
networkRxSpeed	float("network_rx_speed")	nullable	L153
networkTxSpeed	float("network_tx_speed")	nullable	L154
createdAt	timestamp("created_at")	.defaultNow()	L155

1.8 Table services (Drizzle Export: *services* — db/schema.ts #L158-L166)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L159
name	varchar("name", { length: 100 })	.notNull().unique()	L160
displayName	varchar("display_name", { length: 255 })	.notNull()	L161
status	mysqlEnum("status", ["running", "stopped", "error"])	.default("stopped")	L162
uptime	bigint("uptime", { mode: "number" })	nullable	L163
version	varchar("version", { length: 50 })	nullable	L164
lastChecked	timestamp("last_checked")	.defaultNow()	L165

1.9 Table generated_images (Drizzle Export: *generatedImages* — db/schema.ts #L168-L249)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L169
prompt	text("prompt")	.notNull()	L170
provider	varchar("provider", { length: 100 })	nullable	L177
modelUsed	varchar("model_used", { length: 255 })	nullable	L181
resolution	varchar("resolution", { length: 50 })	nullable	L185
steps	int("steps")	nullable	L189
cfg	float("cfg")	nullable	L190
denoise	float("denoise")	nullable	L191
batchSize	int("batch_size")	nullable	L194
sampler	varchar("sampler", { length: 100 })	nullable	L197
scheduler	varchar("scheduler", { length: 100 })	nullable	L201
seed	bigint("seed", { mode: "number", })	nullable	L205
generationTime	int("generation_time")	nullable	L211
gpuUsage	float("gpu_usage")	nullable	L214
vramUsage	float("vram_usage")	nullable	L217
imageUrl	varchar("image_url", { length: 500 })	nullable	L220
baseImageUrl	varchar("base_image_url", { length: 500 })	nullable	L224
upscaledImageUrl	varchar("upscaled_image_url", { length: 500 })	nullable	L228
pipelineType	varchar("pipeline_type", { length: 50 })	nullable	L232
hiresSteps	int("hires_steps")	nullable	L236
hiresDenoise	float("hires_denoise")	nullable	L239
upscaleModel	varchar("upscale_model", { length: 100 })	nullable	L242
createdAt	timestamp("created_at")	.defaultNow()	L246

1.10 Table generated_videos (Drizzle Export: *generatedVideos* — db/schema.ts #L257-L291)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L258
prompt	text("prompt")	.notNull()	L259
negativePrompt	text("negative_prompt")	nullable	L260
provider	varchar("provider", { length: 100 })	nullable	L264
modelUsed	varchar("model_used", { length: 255 })	nullable	L265
resolution	varchar("resolution", { length: 50 })	nullable	L266
steps	int("steps")	nullable	L267
cfg	float("cfg")	nullable	L268
sampler	varchar("sampler", { length: 100 })	nullable	L269
scheduler	varchar("scheduler", { length: 100 })	nullable	L270
seed	bigint("seed", { mode: "number" })	nullable	L271
frameRate	int("frame_rate")	nullable	L274
length	int("length")	nullable	L275
fps	int("fps")	nullable	L276
durationSeconds	float("duration_seconds")	nullable	L277
format	mysqlEnum("format", ["webp", "mp4"])	.default("webp")	L278
sceneCount	int("scene_count")	.default(1)	L282
scenes	json("scenes")	nullable	L283
planningModel	varchar("planning_model", { length: 100 })	nullable	L284
generationTime	int("generation_time")	nullable	L286
gpuUsage	float("gpu_usage")	nullable	L287
vramUsage	float("vram_usage")	nullable	L288
videoUrl	varchar("video_url", { length: 500 })	nullable	L289
createdAt	timestamp("created_at")	.defaultNow()	L290

1.11 Table file_folders (Drizzle Export: *fileFolders* — db/schema.ts #L293-L298)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L294
name	varchar("name", { length: 255 })	.notNull()	L295
parentId	bigint("parent_id", { mode: "number", unsigned: true })	nullable	L296
createdAt	timestamp("created_at")	.defaultNow()	L297

1.12 Table files (Drizzle Export: *files* — db/schema.ts #L300-L309)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L301
name	varchar("name", { length: 255 })	.notNull()	L302
path	varchar("path", { length: 500 })	.notNull()	L303
size	bigint("size", { mode: "number" })	nullable	L304
mimeType	varchar("mime_type", { length: 100 })	nullable	L305
folderId	bigint("folder_id", { mode: "number", unsigned: true })	.references(() => fileFolders.id)	L306
isIndexed	boolean("is_indexed")	.default(false)	L307
createdAt	timestamp("created_at")	.defaultNow()	L308

1.13 Table chat_attachments (Drizzle Export: *chatAttachments* — db/schema.ts #L311-L350)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L312
conversationId	bigint("conversation_id", { mode: "number", unsigned: true, })	.notNull()	L314

1.14 Table research_collections (Drizzle Export: *researchCollections* — db/schema.ts #L352-L357)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L353
name	varchar("name", { length: 255 })	.notNull()	L354
description	text("description")	nullable	L355
createdAt	timestamp("created_at")	.defaultNow()	L356

1.15 Table research_documents (Drizzle Export: *researchDocuments* — db/schema.ts #L359-L368)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L360
title	varchar("title", { length: 500 })	.notNull()	L361
content	text("content")	nullable	L362
source	varchar("source", { length: 500 })	nullable	L363
collectionId	bigint("collection_id", { mode: "number", unsigned: true })	.references(() => researchCollections.id)	L364
isBookmarked	boolean("is_bookmarked")	.default(false)	L366
createdAt	timestamp("created_at")	.defaultNow()	L367

1.16 Table knowledge_entries (Drizzle Export: *knowledgeEntries* — db/schema.ts #L370-L378)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L371
title	varchar("title", { length: 500 })	.notNull()	L372
content	text("content")	.notNull()	L373
sourceType	mysqlEnum("source_type", ["conversation", "research", "file", "manual"])	nullable	L374
tags	json("tags")	nullable	L375
relevanceScore	float("relevance_score")	nullable	L376
createdAt	timestamp("created_at")	.defaultNow()	L377

1.17 Table workflows (Drizzle Export: *workflows* — db/schema.ts #L380-L388)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L381
name	varchar("name", { length: 255 })	.notNull()	L382
description	text("description")	nullable	L383
nodes	json("nodes")	.notNull()	L384
edges	json("edges")	.notNull()	L385
isTemplate	boolean("is_template")	.default(false)	L386
createdAt	timestamp("created_at")	.defaultNow()	L387

1.18 Table activity_logs (Drizzle Export: *activityLogs* — db/schema.ts #L390-L397)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L391
action	varchar("action", { length: 255 })	.notNull()	L392
entityType	varchar("entity_type", { length: 100 })	nullable	L393
entityId	bigint("entity_id", { mode: "number" })	nullable	L394
metadata	json("metadata")	nullable	L395
createdAt	timestamp("created_at")	.defaultNow()	L396

1.19 Table chat_attachment_chunks (Drizzle Export: *chatAttachmentChunks* — db/schema.ts #L591-L613)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L595
attachmentId	bigint("attachment_id", { mode: "number", unsigned: true, })	.notNull()	L597

1.20 Table chat_attachment_embeddings (Drizzle Export: *chatAttachmentEmbeddings* — db/schema.ts #L615-L631)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L619
chunkId	bigint("chunk_id", { mode: "number", unsigned: true, })	.notNull()	L621

1.21 Table voice_conversations (Drizzle Export: *voiceConversations* — db/schema.ts #L633-L659)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L634
title	varchar("title", { length: 255 })	nullable	L636
modelId	varchar("model_id", { length: 100 })	.notNull()	L638
mode	mysqlEnum("mode", ["tts", "stt", "voice"])	.notNull()	L640
providerId	varchar("provider_id", { length: 100 })	nullable	L646
speechModelId	varchar("speech_model_id", { length: 100 })	nullable	L648
voiceId	varchar("voice_id", { length: 100 })	nullable	L650
summary	text("summary")	nullable	L652
summarizedMessageCount	int("summarized_message_count")	.notNull().default(0)	L654
createdAt	timestamp("created_at")	.defaultNow()	L656
updatedAt	timestamp("updated_at")	.defaultNow()	L658

1.22 Table voice_messages (Drizzle Export: *voiceMessages* — db/schema.ts #L661-L747)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L663

1.23 Table document_analysis (Drizzle Export: *documentAnalysis* — db/schema.ts #L754-L779)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	serial("id")	.primaryKey()	L755
fileId	bigint("file_id", { mode: "number", unsigned: true })	.notNull() .unique() .references(() => files.id)	L756
summary	text("summary")	nullable	L760
entities	json("entities")	nullable	L761
keywords	json("keywords")	nullable	L762
topics	json("topics")	nullable	L763
language	varchar("language", { length: 50 })	nullable	L764
documentType	varchar("document_type", { length: 100 })	nullable	L765
confidence	float("confidence")	nullable	L766
status	mysqlEnum("status", ["pending", "ingesting", "summarizing", "extracting", "synthesizing", "complete", "error",])	.default("pending")	L767
error	text("error")	nullable	L776
createdAt	timestamp("created_at")	.defaultNow()	L777
updatedAt	timestamp("updated_at")	.defaultNow()	L778

1.24 Table forge_sessions (Drizzle Export: *forgeSessions* — db/schema.ts #L782-L812)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L783
goal	text("goal")	.notNull()	L784
stackProfileId	varchar("stack_profile_id", { length: 64 })	.notNull()	L785
customStack	text("custom_stack")	nullable	L789
modelId	varchar("model_id", { length: 128 })	.notNull()	L792
workspacePath	varchar("workspace_path", { length: 512 })	.notNull()	L793
allocatedPort	int("allocated_port")	.notNull()	L797
status	mysqlEnum("status", ["planning", "running", "paused", "blocked", "done", "failed"])	.notNull() .default("planning")	L798
sharedContracts	json("shared_contracts")	.notNull()	L808
iterationCount	int("iteration_count")	.notNull().default(0)	L809
createdAt	timestamp("created_at")	.notNull().defaultNow()	L810
updatedAt	timestamp("updated_at")	.notNull().defaultNow()	L811

1.25 Table forge_task_nodes (Drizzle Export: *forgeTaskNodes* — db/schema.ts #L814-L830)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L815
sessionId	varchar("session_id", { length: 36 })	.notNull()	L816
parentId	varchar("parent_id", { length: 36 })	nullable	L817
description	text("description")	.notNull()	L818
acceptanceCriteria	text("acceptance_criteria")	.notNull()	L819
status	mysqlEnum("status", ["pending", "in_progress", "done", "failed", "blocked"])	.notNull() .default("pending")	L820
dependsOn	json("depends_on")	.notNull()	L825
attempts	int("attempts")	.notNull().default(0)	L826
lastError	text("last_error")	nullable	L827
isIntegrationCheck	boolean("is_integration_check")	.notNull().default(false)	L828
createdAt	timestamp("created_at")	.notNull().defaultNow()	L829

1.26 Table forge_iterations (Drizzle Export: *forgeIterations* — db/schema.ts #L832-L840)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L833
sessionId	varchar("session_id", { length: 36 })	.notNull()	L834
taskId	varchar("task_id", { length: 36 })	.notNull()	L835
phase	mysqlEnum("phase", ["think", "act", "evaluate"])	.notNull()	L836
input	text("input")	.notNull()	L837
output	text("output")	.notNull()	L838
timestamp	timestamp("timestamp")	.notNull().defaultNow()	L839

1.27 Table forgex_sessions (Drizzle Export: *forgexSessions* — db/schema.ts #L848-L864)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L849
goal	text("goal")	.notNull()	L850
modelId	varchar("model_id", { length: 128 })	.notNull()	L851
workspacePath	varchar("workspace_path", { length: 512 })	.notNull()	L852
status	mysqlEnum("status", ["starting", "running", "idle", "exited", "failed"])	.notNull() .default("starting")	L853
pid	int("pid")	nullable	L856
exitCode	int("exit_code")	nullable	L857
claudeSessionId	varchar("claude_session_id", { length: 128 })	nullable	L861
createdAt	timestamp("created_at")	.notNull().defaultNow()	L862
updatedAt	timestamp("updated_at")	.notNull().defaultNow()	L863

1.28 Table forgex_output (Drizzle Export: *forgexOutput* — db/schema.ts #L866-L872)

Column Property	Drizzle / SQL Type Definition	Constraints / Default Chain	Line
id	varchar("id", { length: 36 })	.primaryKey()	L867
sessionId	varchar("session_id", { length: 36 })	.notNull()	L868
stream	mysqlEnum("stream", ["stdout", "stderr", "system"])	.notNull()	L869
text	text("text")	.notNull()	L870
timestamp	timestamp("timestamp")	.notNull().defaultNow()	L871

2. Complete Frontend File, Function, Class & Endpoint Inventory (app/frontend/*)

Frontend File Path	Size / Lines	Functions, Classes & API Endpoints (with Exact Line Numbers)
frontend/assets/desktop-only.css	68 lines (1,431 B)	Static HTML / CSS stylesheet asset
frontend/assets/desktop-only.js	43 lines (1,570 B)	Sym: isPhoneDevice() (L18), Sym: apply() (L36)
frontend/assets/liquid-select.css	187 lines (4,617 B)	Static HTML / CSS stylesheet asset
frontend/assets/liquid-select.js	198 lines (6,809 B)	Sym: initLiquidSelect(nativeSelect) (L30), Sym: renderOptions() (L66), Sym: processOption(opt) (L72), Sym: toggleElevation(wrap, elevate) (L139)
frontend/assets/nav.css	91 lines (5,806 B)	Static HTML / CSS stylesheet asset
frontend/assets/nav.js	53 lines (1,494 B)	Sym: handleScroll() (L28)
frontend/assets/page-transitions.css	112 lines (2,845 B)	Static HTML / CSS stylesheet asset
frontend/assets/page-transitions.js	139 lines (4,638 B)	Sym: _clearTimers() (L26), Sym: showBar() (L33), Sym: completeBar() (L53)
frontend/chatspace/chat.css	569 lines (36,662 B)	Static HTML / CSS stylesheet asset
frontend/chatspace/chat.js	1489 lines (51,236 B)	Sym: createThinkingTimer(summaryEl, label = 'Thinking') (L10), Sym: updateTimer() (L15), Sym: isMobile() (L55), Sym: checkSidebarBackdrop() (L89), Sym: showCustomModal({ title, message, type, defaultValue, confirmText, isDanger }) (L111), Sym: closeCustomModal(value = false) (L140), Sym: handleSend() (L200), Sym: generateRealResponse(prompt, uploadedAttachments = []) (L278), Sym: processLine(line) (L338), Sym: safeProcessLine(line) (L390), Sym: generateAgentResponse(prompt, uploadedAttachments = []) (L460), Sym: startRoundBlock(roundNumber) (L510), Sym: processLine(line) (L560), Sym: safeProcessLine(line) (L717), Sym: appendUserMessage(text, files = []) (L781), Sym: appendAIMessage(text, thinking = null, sources = []) (L816), Sym: appendExtendedAIMessage(message) (L869), Sym: renderSourcesPanel(containerId, sources) (L943)
frontend/chatspace/chat.js (cont.)	—	Sym: scrollToBottom() (L970), Sym: setActiveConversation(id) (L974), Sym: escapeHTML(str) (L987), Sym: formatFileSize(bytes) (L998), Sym: getFileIcon(file) (L1004), Sym: renderAttachments() (L1028), Sym: addAttachments(files) (L1061), Sym: createConversation(firstPrompt) (L1135), Sym: saveMessage(role, content, thinking) (L1151), Sym: saveExtendedMessage({ content, thinking, executionSummary, toolCalls }) (L1177), Sym: loadModels() (L1203), Sym: loadKnowledgeFiles() (L1240), Sym: loadConversations() (L1279), Sym: loadConversation(id) (L1342), Sym: deleteConversation(id, event) (L1386), Sym: renameConversation(id, currentTitle) (L1423), Sym: handleNewChat() (L1451), API: /api/chat/stream (L317)
frontend/chatspace/chat.js (cont.)	—	API: /api/chat/sources (L431), API: /api/chat/agent-stream (L544), API: /api/chat/sources (L767), API: /api/trpc/conversation.create (L1136), API: /api/trpc/message.create (L1154), API: /api/trpc/message.createWithToolCalls (L1181), API: /api/trpc/model.list (L1205), API: /api/trpc/file.list (L1242), API: /api/trpc/conversation.list (L1281), API: /api/trpc/conversation.delete (L1403), API: /api/trpc/conversation.update (L1435)
frontend/chatspace/index.html	265 lines (14,479 B)	Static HTML / CSS stylesheet asset
frontend/files/files.css	332 lines (26,041 B)	Static HTML / CSS stylesheet asset

Frontend File Path	Size / Lines	Functions, Classes & API Endpoints (with Exact Line Numbers)
`frontend/files/files.js`	871 lines (34,526 B)	Sym: isMobile() (L18), Sym: checkSidebarBackdrop() (L37), Sym: formatBytes(bytes, decimals = 2) (L61), Sym: formatDate(dateStr) (L70), Sym: escapeHTML(str) (L75), Sym: getIconForMime(mimeType) (L86), Sym: loadFolders() (L95), Sym: renderFolders() (L108), Sym: loadFiles(query = "") (L201), Sym: renderFiles() (L234), Sym: loadExtractedText(fileId) (L470), Sym: analysisToastContainer() (L546), Sym: renderToast(fileId, fileName, run) (L550), Sym: renderAnalysisProgress(run) (L683), Sym: buildAnalysisHTML(analysis) (L740), Sym: displayValue(v) (L749), Sym: entitySection(label, list) (L764), Sym: loadAnalysisTab(fileId) (L797)
`frontend/files/files.js (cont.)`	—	API: /api/trpc/file.listFolders (L97), API: /api/trpc/file.createFolder (L162), API: /api/trpc/file.deleteFolder (L181), API: /api/files/upload (L318), API: /api/trpc/file.delete (L353), API: /api/trpc/fileContent.getByFileId?input=${payload} (L482), API: /api/files/${fileId}/analyze-stream (L604), API: /api/files/${fileId}/analysis (L815)
`frontend/files/index.html`	251 lines (13,394 B)	Static HTML / CSS stylesheet asset
`frontend/forge/forge.css`	321 lines (21,039 B)	Static HTML / CSS stylesheet asset
`frontend/forge/forge.js`	811 lines (31,172 B)	Sym: resize() (L29), Sym: tick() (L57), Sym: resize() (L96), Sym: tick() (L114), Sym: resetRawActivityPanel() (L170), Sym: setRawStatusDot(active, text) (L177), Sym: updateRawActivityStatus(payload) (L184), Sym: startNewRawStreamRun() (L198), Sym: streamRawText(text) (L215), Sym: appendThinkText(body, text) (L257), Sym: appendFinalText(body, text) (L267), Sym: refreshSessionList() (L327), Sym: statusDotClass(status) (L361), Sym: selectSession(sessionId) (L369), Sym: updatePauseResumeButton(status) (L389), Sym: updateFollowupBar(status) (L410), Sym: togglePauseResume(action) (L421), Sym: updateEngineStatus(status) (L437)
`frontend/forge/forge.js (cont.)`	—	Sym: loadTaskTree(sessionId) (L445), Sym: renderTaskTree(tasks) (L456), Sym: focusTask(taskId) (L494), Sym: connectLiveLog(sessionId) (L509), Sym: summarizeOutput(output) (L630), Sym: appendLogEvent(evt) (L648), Sym: renderLogGroup(taskId, events, { replace, expanded }) (L652), Sym: sendFollowUp() (L768), Sym: escapeHtml(str) (L801), API: ${this.base}/forge.${proc} (L301)
`frontend/forge/index.html`	292 lines (14,372 B)	Static HTML / CSS stylesheet asset
`frontend/forgex/forgex.css`	321 lines (21,359 B)	Static HTML / CSS stylesheet asset
`frontend/forgex/forgex.js`	360 lines (14,618 B)	Sym: resize() (L23), Sym: tick() (L38), Sym: resize() (L67), Sym: tick() (L81), Sym: refreshSessionList() (L135), Sym: statusDotClass(status) (L166), Sym: selectSession(sessionId) (L175), Sym: updateEngineStatus(status) (L192), Sym: updateFollowupBar(status) (L213), Sym: connectLiveTerminal(sessionId) (L237), Sym: appendTerminalLine(stream, text) (L268), Sym: sendFollowUp() (L332), Sym: escapeHtml(str) (L352), API: ${this.base}/forgex.${proc} (L112)
`frontend/forgex/index.html`	223 lines (11,038 B)	Static HTML / CSS stylesheet asset
`frontend/framesx/framesx.css`	953 lines (28,691 B)	Static HTML / CSS stylesheet asset
`frontend/framesx/framesx.js`	942 lines (35,689 B)	Sym: snapTo32(value, min, max) (L203), Sym: checkSceneLogic() (L224), Sym: aspectToDimensions(aspect) (L235), Sym: loadPlanningModels() (L250), Sym: renderGroup(title, groupModels) (L265), Sym: loadVideoProviders() (L295), Sym: applyProvider(p) (L328), Sym: startGeneration() (L365), Sym: setupSceneTracker(sceneCount) (L444), Sym: updateSceneDots(currentIndex, total, allDone) (L459), Sym: logLine(text, tagClass) (L473), Sym: logToken(token) (L486), Sym: resetTokenLine() (L495), Sym: connectLiveStream(jobId) (L500), Sym: startStatusPolling(jobId) (L601), Sym: stopStatusPolling() (L632), Sym: showResult(result) (L640), Sym: loadGallery() (L687)
`frontend/framesx/framesx.js (cont.)`	—	Sym: buildGalleryItem(video) (L749), Sym: reuseVideoSettings(video) (L815), Sym: openVideoModal(video) (L903), Sym: closeVideoModal() (L933), API: /api/models/list (L253), API: /api/video/generate (L417), API: /api/video/stream/${jobId} (L505), API: /api/video/status/${jobId} (L605), API: /api/video/result/${jobId} (L617), API: /api/video/list (L692), API: /api/video/${video.id} (L789)
`frontend/framesx/index.html`	464 lines (26,143 B)	Static HTML / CSS stylesheet asset

Frontend File Path	Size / Lines	Functions, Classes & API Endpoints (with Exact Line Numbers)
frontend/home/index.html	317 lines (17,030 B)	Static HTML / CSS stylesheet asset
frontend/home/script.js	159 lines (5,286 B)	Sym: resize() (L36), Sym: tick() (L52), Sym: loadImages() (L90), Sym: tick(t) (L149), API: /api/trpc/image.list (L92)
frontend/home/styles.css	386 lines (24,424 B)	Static HTML / CSS stylesheet asset
frontend/model/index.html	122 lines (5,879 B)	Static HTML / CSS stylesheet asset
frontend/model/model-manager.css	199 lines (12,856 B)	Static HTML / CSS stylesheet asset
frontend/model/model-manager.js	426 lines (17,189 B)	Sym: class EventManager (L9), Sym: class ApiManager (L51), Sym: class StateManager (L123), Sym: class ComponentFactory (L157), Sym: class FilterManager (L236), Sym: class OverviewManager (L276), Sym: class PackageManager (L293), Sym: class ActionManager (L319), Sym: class DownloadManager (L350), Sym: class App (L403)
frontend/monitor/index.html	332 lines (15,310 B)	Static HTML / CSS stylesheet asset
frontend/monitor/monitor.css	291 lines (16,687 B)	Static HTML / CSS stylesheet asset
frontend/monitor/monitor.js	323 lines (8,557 B)	Sym: initSparkline(id, bars) (L39), Sym: updateSparkline(id) (L48), Sym: updateSystemStats() (L66), Sym: updateServices() (L138), Sym: updateProcesses() (L188), Sym: updateStorage() (L234), Sym: addEvent() (L296), API: /api/system/current (L68), API: /api/system/services (L141), API: /api/system/processes (L192), API: /api/system/storage (L238)
frontend/personastudio/index.html	168 lines (8,373 B)	Static HTML / CSS stylesheet asset
frontend/personastudio/personastudio.css	199 lines (14,643 B)	Static HTML / CSS stylesheet asset
frontend/personastudio/personastudio.js	371 lines (13,770 B)	Sym: trpcQuery(path, input) (L4), Sym: trpcMutation(path, input) (L15), Sym: init() (L43), Sym: startPollingIfNeeded() (L67), Sym: pollState() (L79), Sym: updateBadge(text, className) (L100), Sym: renderServerState() (L105), Sym: renderLogConsole(logArray) (L215), Sym: scrollToBottomOfLogs() (L221), Sym: populateVoices() (L271), Sym: renderPersonas() (L281), Sym: escapeHtml(unsafe) (L349)
frontend/robotics/index.html	345 lines (17,391 B)	Static HTML / CSS stylesheet asset
frontend/robotics/robotics.css	235 lines (15,697 B)	Static HTML / CSS stylesheet asset
frontend/robotics/robotics.js	81 lines (2,985 B)	Sym: updateTopologyLines() (L27)
frontend/settings/index.html	531 lines (29,182 B)	Static HTML / CSS stylesheet asset
frontend/settings/settings.css	342 lines (20,498 B)	Static HTML / CSS stylesheet asset
frontend/settings/settings.js	581 lines (21,978 B)	Sym: trpcQuery(path, input) (L22), Sym: trpcMutation(path, input) (L33), Sym: runAction(btn, fn) (L62), Sym: fmtBytes(bytes) (L85), Sym: loadModels() (L165), Sym: renderActiveModelCard(m) (L190), Sym: renderInstalledModelRow(m) (L208), Sym: wireModelButtons() (L222), Sym: loadPython() (L239), Sym: refreshPythonStatus() (L255), Sym: loadSpeech() (L271), Sym: refreshSpeechStatus() (L288), Sym: loadSpeechProviders() (L307), Sym: loadCluster() (L327), Sym: renderWorkerCard(w) (L341), Sym: wireWorkerCardClicks() (L360), Sym: fmtRate(bytesPerSec) (L376), Sym: closeWorkerModal() (L382)
frontend/settings/settings.js (cont.)	—	Sym: openWorkerModal(workerId) (L403), Sym: refreshWorkerMetrics() (L432), Sym: renderMetricSeries(points) (L442), Sym: drawSparkline(canvasId, values, min, max) (L465), Sym: startLogStream() (L506), Sym: appendLogLine(entry) (L548), Sym: escapeHtml(str) (L564)

Frontend File Path	Size / Lines	Functions, Classes & API Endpoints (with Exact Line Numbers)
`frontend/studio/index.html`	478 lines (27,037 B)	Static HTML / CSS stylesheet asset
`frontend/studio/studio.css`	896 lines (26,513 B)	Static HTML / CSS stylesheet asset
`frontend/studio/studio.js`	1342 lines (42,638 B)	Sym: isMobile() (L34), Sym: checkSidebarBackdrop() (L83), Sym: parseAspectRatio(str) (L152), Sym: gcd(a, b) (L166), Sym: formatRatio(w, h) (L170), Sym: updateAspectRatioUI(w, h, ratioStr, source) (L198), Sym: getSelectedResolution() (L344), Sym: getGenerationConfig() (L353), Sym: getUpscaleConfig() (L377), Sym: connectLiveStream(jobId) (L448), Sym: pollGeneration(jobId) (L551), Sym: loadResult(jobId) (L584), Sym: loadGallery() (L657), Sym: renderGallery() (L683), Sym: showImage(id) (L834), Sym: updateMetadata(image) (L889), Sym: updateModalUI(image) (L917), Sym: openImageModal(id) (L957)
`frontend/studio/studio.js (cont.)`	—	Sym: closeImageModal() (L975), Sym: reuseConfiguration() (L988), Sym: updateSliderLabels() (L1083), Sym: previousImage() (L1124), Sym: nextImage() (L1130), Sym: downloadImage(image) (L1136), Sym: copyPrompt(image) (L1143), Sym: loadImageProviders() (L1171), Sym: applyProviderDefaults(provider) (L1214), Sym: applyUpscaleDefaults(provider) (L1229), Sym: loadComfyOptions() (L1293), Sym: populateSelect(id, items, currentVal) (L1301), API: /api/image/generate (L417), API: /api/image/stream/${jobId} (L453), API: /api/image/status/${jobId} (L554), API: /api/image/result/${jobId} (L586), API: /api/trpc/image.list (L659)
`frontend/voice/index.html`	343 lines (19,504 B)	Static HTML / CSS stylesheet asset
`frontend/voice/voice.css`	557 lines (33,601 B)	Static HTML / CSS stylesheet asset
`frontend/voice/voice.js`	2226 lines (92,031 B)	Sym: class VoiceApi (L47), Sym: class SettingsManager (L158), Sym: class DeviceManager (L175), Sym: class AudioEngine (L205), Sym: class AudioQueue (L340), Sym: class HistoryManager (L420), Sym: class Visualizer (L517), Sym: class StateManager (L772), Sym: class VoiceUI (L783), Sym: openSettings() (L837), Sym: closeSettings() (L841), Sym: renderGroup(title, groupModels) (L1096), Sym: tick() (L1163), Sym: class VoiceConversation (L1854), Sym: processLine(line) (L2049), Sym: safeProcessLine(line) (L2112), API: ${this.base}${endpoint} (L51), API: ${this.base}/profiles (L66)
`frontend/voice/voice.js (cont.)`	—	API: /api/voice/conversation (L74), API: ${this.base}/tts (L104), API: ${this.base}/stt (L114), API: /api/chat/stream (L123), API: /api/trpc/model.list (L131), API: /api/voice/conversation/${id} (L150), API: /api/voice/conversations (L1363)

3. Complete Backend TypeScript File, Export & Route Inventory (api/, db/, worker/)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/boot.ts	2630 lines	[Route] GET /api/system/current (L151), [Route] POST /api/chat (L159), [Route] POST /api/chat/stream (L241), [Route] POST /api/chat/agent-stream (L1212), [Route] GET /api/speech/providers (L1385), [Route] GET /api/speech/models (L1398), [Route] GET /api/speech/voices (L1437), [Route] GET /api/speech/profiles (L1486), [Route] POST /api/speech/tts (L1499), [Route] POST /api/speech/stt (L1596), [Route] PUT /api/speech/profiles (L1717), [Route] GET /api/speech/health (L1764), [Route] POST /api/voice/conversation (L1775), [Route] POST /api/voice/message (L1807), [Route] POST /api/voice/conversation/rename (L1864), [Route] GET /api/voice/messages/:id (L1933), [Route] GET /api/voice/audio/:id (L1951), [Route] GET /api/voice/conversations (L1973)
app/api/boot.ts (cont.)	—	[Route] GET /api/voice/conversation/:id (L1986), [Route] GET /api/voice/messages (L2032), [Route] GET /api/system/services (L2053), [Route] GET /api/system/processes (L2070), [Route] GET /api/system/storage (L2100), [Route] GET /api/providers (L2114), [Route] POST /api/image/generate (L2136), [Route] GET /api/image/stream/:jobId (L2193), [Route] GET /api/image/status/:jobId (L2250), [Route] GET /api/image/result/:jobId (L2257), [Route] POST /api/video/generate (L2299), [Route] GET /api/video/status/:jobId (L2334), [Route] GET /api/video/result/:jobId (L2341), [Route] GET /api/video/stream/:jobId (L2358), [Route] GET /api/video/list (L2421), [Route] DELETE /api/video/:id (L2430), [Route] GET /api/models/list (L2464), [Route] GET /uploads/* (L2504)
app/api/boot.ts (cont.)	—	[Route] GET /api/test-embedding (L2510), [Route] GET /api/backfill-embeddings (L2515), [Route] GET /api/search (L2520), [Route] GET /api/chat/sources (L2527), [Route] GET /api/test-tavily (L2531), [Route] POST /api/cluster/register (L2536), [Route] POST /api/cluster/heartbeat (L2545), [Route] GET /api/cluster/workers (L2561), [Route] GET /api/cluster/:id/logs/stream (L2569)
app/api/context.ts	12 lines	[Export] type TrpcContext (L3), [Export] async fn createContext (L8)
app/api/middleware.ts	10 lines	[Export] const createRouter (L9), [Export] const publicQuery (L10)
app/api/router.ts	47 lines	[Route] tRPC query: ping (L24), [Export] const appRouter (L23), [Export] type AppRouter (L47)
app/api/forge/agentLoop.ts	581 lines	[Export] type IterationResult (L30), [Export] async fn writeTodoFile (L379), [Export] async fn runIteration (L422)
app/api/forge/constants.ts	51 lines	[Export] const FORGE_ROOT (L12), [Export] const FORGE_MODEL (L20), [Export] const OLLAMA_TAGS_URL (L29), [Export] const OLLAMA_GENERATE_URL (L30), [Export] const DEFAULT_COMMAND_TIMEOUT_MS (L32), [Export] const INSTALL_COMMAND_TIMEOUT_MS (L36), [Export] const DEFAULT_MODEL_TIMEOUT_MS (L40), [Export] const MAX_ATTEMPTS_PER_TASK (L42), [Export] const MAX_INTEGRATION_CHECK_ATTEMPTS (L49), [Export] const MAX_CONSECUTIVE_IDENTICAL_ERRORS (L51)
app/api/forge/contracts.ts	70 lines	[Export] async fn registerContract (L17), [Export] async fn getContractsForTask (L32), [Export] fn contractKeyForTask (L68)
app/api/forge/db.ts	408 lines	[Export] fn findFreePort (L104), [Export] async fn createSession (L122), [Export] async fn getSession (L154), [Export] async fn listSessions (L167), [Export] async fn getAllSessions (L178), [Export] async fn updateSessionStatus (L187), [Export] async fn updateSharedContracts (L200), [Export] async fn incrementSessionIterationCount (L216), [Export] async fn createTaskNode (L230), [Export] async fn getTask (L263), [Export] async fn getTaskTree (L273), [Export] async fn updateTaskStatus (L283), [Export] async fn incrementTaskAttempts (L299), [Export] async fn getNextActionableTask (L313), [Export] async fn logIteration (L337), [Export] async fn getIterationsForTask (L362), [Export] async fn getIterationsSince (L388)
app/api/forge/execWorker.ts	237 lines	[Export] class ForgePathViolation (L22), [Export] class ForgeCommandViolation (L29), [Export] fn resolveSafePath (L40), [Export] fn validateCommand (L72), [Export] type CommandResult (L92), [Export] async fn runCommand (L102), [Export] fn setActiveTask (L160), [Export] async fn writeFile (L170), [Export] async fn readFile (L188), [Export] async fn deleteFile (L205), [Export] async fn listDir (L221)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/forge/modelClient.ts	290 lines	[Export] class ForgeModelError (L19), [Export] fn getInFlightCall (L37), [Export] fn subscribeToTokens (L64), [Export] fn getTokenBuffer (L72), [Export] async fn callForgeModel (L98), [Export] fn extractJson (L230)
app/api/forge/orchestrator.ts	229 lines	[Export] async fn createSession (L19), [Export] async fn pauseSession (L130), [Export] async fn resumeSession (L135), [Export] async fn addFollowUp (L157), [Export] async fn driveSessionToCompletion (L208)
app/api/forge/router.ts	180 lines	[Route] tRPC query: list (L59), [Route] tRPC query: listModels (L73), [Route] tRPC query: listStacks (L133), [Export] const forgeRouter (L18)
app/api/forge/routes.ts	169 lines	[Route] GET /:sessionId/stream (L38)
app/api/forge/schema.ts	92 lines	[Export] const forgeSessions (L29), [Export] const forgeTaskNodes (L62), [Export] const forgeIterations (L84)
app/api/forge/taskTree.ts	140 lines	[Export] async fn pickNextTask (L31), [Export] async fn checkStuck (L88), [Export] async fn markStuckAndContinue (L127)
app/api/forge/types.ts	126 lines	[Export] type ForgeSessionStatus (L10), [Export] type TaskNodeStatus (L18), [Export] type ForgeIterationPhase (L25), [Export] type ForgeSession (L27), [Export] type TaskNode (L47), [Export] type ForgeIteration (L61), [Export] type AgentAction (L80), [Export] type AgentActionType (L117), [Export] type EvaluationResult (L123)
app/api/forge/prompts/evaluate.ts	65 lines	[Export] fn buildEvaluatePrompt (L11)
app/api/forge/prompts/integrationCheck.ts	59 lines	[Export] fn buildIntegrationCheckPrompt (L13)
app/api/forge/prompts/planning.ts	96 lines	[Export] type PlannedTask (L13), [Export] fn buildPlanningPrompt (L19)
app/api/forge/prompts/think.ts	113 lines	[Export] fn buildThinkPrompt (L13)
app/api/forge/stacks/general.ts	118 lines	[Export] const general (L23)
app/api/forge/stacks/javaSpringBoot.ts	111 lines	[Export] const javaSpringBoot (L11)
app/api/forge/stacks/mernStack.ts	133 lines	[Export] const mernStack (L13)
app/api/forge/stacks/nodeExpress.ts	156 lines	[Export] const nodeExpress (L13)
app/api/forge/stacks/pythonFastapi.ts	162 lines	[Export] const pythonFastapi (L11)
app/api/forge/stacks/registry.ts	36 lines	[Export] fn getStackProfile (L24), [Export] fn listStackProfiles (L34)
app/api/forge/stacks/types.ts	30 lines	[Export] type TaskTemplate (L10), [Export] type StackProfile (L20)
app/api/forgex/constants.ts	58 lines	[Export] const FORGEX_ROOT (L18), [Export] const OLLAMA_ANTHROPIC_BASE_URL (L26), [Export] const OLLAMA_TAGS_URL (L27), [Export] const CLAUDE_CODE_BIN (L33), [Export] const RECOMMENDED_MIN_CONTEXT_TOKENS (L41), [Export] const PROCESS_STARTUP_TIMEOUT_MS (L49), [Export] const OUTPUT_BUFFER_MAX_LINES (L54), [Export] fn workspacePathFor (L56)
app/api/forgex/db.ts	142 lines	[Export] async fn createSession (L40), [Export] async fn getSession (L66), [Export] async fn listSessions (L76), [Export] async fn updateSessionStatus (L85), [Export] async fn logOutputLine (L103), [Export] async fn getOutputSince (L122)
app/api/forgex/processManager.ts	611 lines	[Export] fn subscribeToOutput (L84), [Export] fn getBufferedOutput (L92), [Export] fn isRunning (L99), [Export] fn subscribeToStatus (L111), [Export] async fn startSession (L548), [Export] async fn sendInput (L566), [Export] async fn stopSession (L596)
app/api/forgex/router.ts	101 lines	[Route] tRPC query: listModels (L38), [Route] tRPC query: list (L64), [Export] const forgexRouter (L14)
app/api/forgex/routes.ts	99 lines	[Route] GET /:sessionId/stream (L20)
app/api/forgex/schema.ts	43 lines	[Export] const forgexSessions (L19), [Export] const forgexOutput (L37)
app/api/forgex/types.ts	46 lines	[Export] type ForgeXSessionStatus (L5), [Export] type ForgeXSession (L12), [Export] type ForgeXOutputLine (L32), [Export] type ForgeXInputMessage (L43)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/lib/env.ts	22 lines	[Export] const env (L11)
app/api/lib/http.ts	77 lines	[Export] class HttpClient (L7)
app/api/lib/temporalContext.ts	110 lines	[Export] interface TemporalContextInput (L25), [Export] fn buildTemporalContext (L38)
app/api/lib/vite.ts	23 lines	[Export] fn serveStaticFiles (L9)
app/api/queries/connection.ts	67 lines	[Export] fn getDb (L51), [Export] async fn getDbAsync (L65)
app/api/queries/conversation.ts	83 lines	[Export] async fn getConversationSummary (L5), [Export] async fn updateConversationSummary (L21), [Export] async fn getConversationSummarizedCount (L44), [Export] async fn updateConversationSummaryWithCount (L63)
app/api/routers/activity.ts	35 lines	[Export] const activityRouter (L7)
app/api/routers/chat.ts	55 lines	[Export] const chatRouter (L8)
app/api/routers/cluster.ts	69 lines	[Route] tRPC query: workers (L13), [Export] const clusterRouter (L12)
app/api/routers/conversation.ts	109 lines	[Route] tRPC query: list (L8), [Export] const conversationRouter (L7)
app/api/routers/file.ts	276 lines	[Export] const fileRouter (L14)
app/api/routers/fileContent.ts	33 lines	[Export] const fileContentRouter (L7)
app/api/routers/image.ts	340 lines	[Route] tRPC query: list (L13), [Export] const imageRouter (L12)
app/api/routers/knowledge.ts	60 lines	[Export] const knowledgeRouter (L7)
app/api/routers/message.ts	211 lines	[Export] const messageRouter (L33)
app/api/routers/model.ts	166 lines	[Route] tRPC query: list (L10), [Route] tRPC query: activeMetrics (L61), [Export] const modelRouter (L9)
app/api/routers/personaplex.ts	400 lines	[Route] tRPC query: voices (L314), [Route] tRPC query: list (L317), [Route] tRPC query: status (L353), [Route] tRPC mutation: start (L363), [Route] tRPC query: status (L375), [Route] tRPC mutation: stop (L395), [Export] const PERSONAPLEX_VOICES (L43), [Export] type Persona (L64), [Export] const personaplexRouter (L312)
app/api/routers/providers.ts	69 lines	[Route] tRPC query: comfyOptions (L33), [Export] const providersRouter (L13)
app/api/routers/research.ts	53 lines	[Route] tRPC query: listCollections (L8), [Export] const researchRouter (L7)
app/api/routers/runtime.ts	26 lines	[Route] tRPC query: status (L12), [Route] tRPC mutation: start (L16), [Route] tRPC mutation: stop (L21), [Export] const runtimeRouter (L10)
app/api/routers/system.ts	137 lines	[Route] tRPC query: currentStats (L11), [Route] tRPC query: storage (L34), [Route] tRPC query: services (L47), [Route] tRPC query: serviceHealth (L51), [Route] tRPC query: processes (L94), [Export] const systemRouter (L10)
app/api/routers/video.ts	110 lines	[Route] tRPC query: list (L11), [Export] const videoRouter (L10)
app/api/routers/workflow.ts	66 lines	[Route] tRPC query: list (L8), [Export] const workflowRouter (L7)
app/api/services/agentLoop.ts	1889 lines	[Export] type AgentEvent (L218), [Export] type AgentEventSink (L237), [Export] async fn synthesizeExecutionSummary (L1170), [Export] interface AgentLoopParams (L1344), [Export] async fn runAgentLoop (L1354)
app/api/services/analyzeStream.ts	198 lines	Internal module / implementation file
app/api/services/backfillEmbeddings.ts	67 lines	Internal module / implementation file
app/api/services/backgroundGeneration.ts	145 lines	[Export] async fn runGeneration (L10)
app/api/services/chatAttachmentSearch.ts	130 lines	[Export] async fn searchChatAttachments (L14)
app/api/services/chatUpload.ts	203 lines	Internal module / implementation file
app/api/services/chunker.ts	19 lines	[Export] fn createChunks (L1)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/services/cluster.ts	282 lines	[Export] interface Worker (L10), [Export] fn registerWorker (L69), [Export] fn heartbeat (L119), [Export] fn removeOfflineWorkers (L135), [Export] fn getWorkers (L158), [Export] fn getWorker (L164), [Export] fn selectWorker (L170), [Export] async fn executeClusterTool (L210)
app/api/services/clusterMetricsHistory.ts	65 lines	[Export] interface MetricPoint (L10), [Export] fn recordMetricPoint (L27), [Export] fn getMetricHistory (L57), [Export] fn clearMetricHistory (L63)
app/api/services/comfy.ts	626 lines	[Export] type ImageGenEvent (L36), [Export] type ImageEmit (L44), [Export] async fn generateFluxImage (L67), [Export] async fn generateLtxVideoPipeline (L467)
app/api/services/comfyWorkflow.ts	125 lines	[Export] interface GenerationConfig (L5), [Export] fn buildWorkflow (L37)
app/api/services/comfyWorkflowFull.ts	264 lines	[Export] interface FullPipelineConfig (L17), [Export] fn buildFullPipelineWorkflow (L111)
app/api/services/companion.ts	820 lines	[Export] fn startCompanionService (L746), [Export] fn stopCompanionService (L810)
app/api/services/contextWindow.ts	209 lines	[Export] type ConversationMessageLike (L35), [Export] const MAX_CONTEXT_TOKENS (L44), [Export] const MAX_CONTEXT_MESSAGES (L47), [Export] fn estimateTokens (L54), [Export] type ContextWindow (L65), [Export] fn computeContextWindow (L82), [Export] type BuiltConversationContext (L118), [Export] async fn buildConversationContext (L144)
app/api/services/conversationMemory.ts	45 lines	[Export] async fn getConversationMemory (L3), [Export] async fn getConversationMessages (L34)
app/api/services/conversationSummary.ts	105 lines	[Export] async fn summarizeConversation (L8)
app/api/services/documentProcessor.ts	76 lines	[Export] interface ProcessedDocument (L5), [Export] async fn processDocument (L35)
app/api/services/embeddingService.ts	37 lines	[Export] async fn generateEmbedding (L1)
app/api/services/fileProcessor.ts	130 lines	[Export] async fn extractText (L14)
app/api/services/firecrawl.ts	90 lines	[Export] fn formatPageContent (L31)
app/api/services/framesx.ts	1036 lines	[Export] type FramesXEvent (L83), [Export] interface VideoGenerationInput (L103), [Export] async fn generateFramesXVideo (L825), [Export] async fn runFramesXGeneration (L978)
app/api/services/generationState.ts	62 lines	[Export] type GenerationJob (L3), [Export] const generationJobs (L37), [Export] fn createJob (L40)
app/api/services/knowledgeWorkflow.ts	143 lines	[Export] interface DocumentAnalysisResult (L6), [Export] interface KnowledgeWorkflowResult (L24), [Export] async fn knowledgeWorkflow (L66)
app/api/services/memory.ts	312 lines	[Export] async fn memoryStore (L21), [Export] async fn memorySearch (L177), [Export] async fn getAllMemories (L188), [Export] async fn memoryUpdate (L199), [Export] async fn memoryDelete (L298)
app/api/services/memoryContext.ts	37 lines	[Export] async fn buildMemoryContext (L3)
app/api/services/memorySemanticSearch.ts	146 lines	[Export] async fn searchMemories (L48)
app/api/services/modelManager.ts	14 lines	Internal module / implementation file
app/api/services/ollamaControl.ts	11 lines	Internal module / implementation file
app/api/services/openRouter.ts	257 lines	[Export] const OPENROUTER_MODELS (L25), [Export] const OPENROUTER_PREFIX (L73), [Export] fn isOpenRouterModel (L76), [Export] fn toOpenRouterId (L84), [Export] fn listOpenRouterModels (L91), [Export] interface OpenRouterMessage (L110), [Export] async fn chatWithOpenRouter (L119), [Export] type OpenRouterToken (L152), [Export] async fn generateWithOpenRouter (L247)
app/api/services/phaseTracker.ts	542 lines	[Export] type PhaseStatus (L26), [Export] type CompletionEvent (L27), [Export] interface ExecutionPhase (L29), [Export] interface ExecutionState (L47), [Export] class PhaseTrackerService (L178)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/services/providers.ts	144 lines	[Export] type MediaType (L14), [Export] type Executor (L15), [Export] interface UpscaleDefaults (L17), [Export] interface ProviderDefinition (L26), [Export] const PROVIDERS (L54), [Export] fn getProvider (L129), [Export] fn listProviders (L133), [Export] fn getDefaultProvider (L138)
app/api/services/pythonTools.ts	17 lines	[Export] const PYTHON_TOOLS (L1)
app/api/services/rag.ts	214 lines	[Export] async fn buildRagPrompt (L5)
app/api/services/research.ts	81 lines	Internal module / implementation file
app/api/services/semanticSearch.ts	307 lines	[Export] async fn searchKnowledge (L92)
app/api/services/sqlQuery.ts	31 lines	Internal module / implementation file
app/api/services/tavily.ts	84 lines	[Export] async fn searchInternet (L3)
app/api/services/telemetry.ts	105 lines	[Export] fn startTelemetryCollector (L17)
app/api/services/toolExecutor.ts	301 lines	[Export] interface ExecutionResult (L35), [Export] async fn executeTool (L64)
app/api/services/toolPattern.ts	509 lines	[Export] interface PatternMatch (L31), [Export] type PatternResolution (L37), [Export] fn matchPatterns (L434), [Export] fn buildDefaultArguments (L495)
app/api/services/toolPipeline.ts	267 lines	[Export] interface ToolHistoryEntry (L96), [Export] interface PipelineResult (L101), [Export] async fn processTools (L174)
app/api/services/toolRouter.ts	827 lines	[Export] type ToolDecision (L50), [Export] const ROUTER_MODEL (L71), [Export] async fn shouldUseTools (L586), [Export] async fn resolveAgentToolCall (L727)
app/api/services/toolSelector.ts	18 lines	[Export] interface ToolCall (L15)
app/api/services/tools.ts	507 lines	[Export] interface ToolDef (L11), [Export] const availableTools (L21), [Export] fn getToolDef (L333), [Export] fn toolsForPrompt (L338), [Export] fn toolsForRouterPrompt (L349), [Export] interface OllamaToolSchema (L402), [Export] fn toolsForNativeSchema (L479)
app/api/services/upload.ts	175 lines	Internal module / implementation file
app/api/services/vectorSearch.ts	39 lines	[Export] fn cosineSimilarity (L1)
app/api/services/videoGenerationState.ts	85 lines	[Export] type SceneStatus (L3), [Export] type VideoJob (L20), [Export] const videoJobs (L51), [Export] fn createVideoJob (L53), [Export] fn recomputeOverallProgress (L74)
app/api/services/videoWorkflow.ts	406 lines	[Export] interface SceneConfig (L5), [Export] fn buildVideoWorkflow (L136), [Export] interface Stage1Config (L230), [Export] interface Stage2Config (L243), [Export] interface Stage3Config (L259), [Export] fn buildStage1Workflow (L320), [Export] fn buildStage2Workflow (L354), [Export] fn buildStage3Workflow (L384)
app/api/services/voiceConversationMemory.ts	125 lines	[Export] async fn getVoiceConversationMemory (L5), [Export] async fn getVoiceConversationMessages (L41), [Export] async fn getVoiceConversationSummary (L63), [Export] async fn getVoiceConversationSummarizedCount (L83), [Export] async fn updateVoiceConversationSummary (L102)
app/api/services/workerClient.ts	227 lines	[Export] async fn executeRemote (L123), [Export] async fn pingWorker (L202)
app/api/services/workerPoller.ts	169 lines	[Export] fn startWorkerPoller (L147)
app/api/services/companion/registry.ts	290 lines	[Export] type DeviceType (L14), [Export] type ConnectionState (L23), [Export] interface DeviceEvent (L25), [Export] interface RuntimeDevice (L31), [Export] interface RegisterInput (L85), [Export] const deviceRegistry (L285)
app/api/services/python-runtime/client.ts	128 lines	[Export] class PythonRuntimeClient (L6), [Export] const pythonRuntimeClient (L127)
app/api/services/python-runtime/index.ts	3 lines	Internal module / implementation file
app/api/services/python-runtime/manager.ts	36 lines	[Export] class PythonRuntimeManager (L3), [Export] const pythonRuntimeManager (L35)
app/api/services/python-runtime/types.ts	28 lines	[Export] interface PythonRuntimeInfo (L1), [Export] interface PythonExecuteRequest (L19), [Export] interface PythonExecuteResponse (L24)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/services/runtime/freePort.ts	79 lines	[Export] fn freePort (L23)
app/api/services/runtime/health.ts	48 lines	[Export] async fn waitForHealth (L1)
app/api/services/runtime/manager.ts	187 lines	[Export] const runtimeManager (L155)
app/api/services/runtime/process.ts	70 lines	[Export] interface RuntimeProcess (L3), [Export] fn startProcess (L13), [Export] fn stopProcess (L56), [Export] fn isRunning (L64)
app/api/services/runtime/python.ts	41 lines	[Export] const PYTHON_PORT (L8), [Export] fn startPythonRuntime (L10)
app/api/services/runtime/speech.ts	40 lines	[Export] const SPEECH_PORT (L8), [Export] fn startSpeechRuntime (L10)
app/api/speech/index.ts	49 lines	[Export] async fn initializeSpeechSystem (L27), [Export] const speechRouter (L35)
app/api/speech/lazy-speech-manager.ts	36 lines	[Export] async fn getSpeechManager (L20), [Export] async fn initializeLazySpeechManager (L32)
app/api/speech/types.ts	440 lines	[Export] type ProviderId (L4), [Export] type ProviderType (L5), [Export] type ModelId (L6), [Export] type VoiceId (L7), [Export] type ProfileId (L8), [Export] type RuntimeId (L9), [Export] type DownloadId (L10), [Export] type DeviceId (L11), [Export] type RequestId (L12), [Export] enum ModelStatus (L15), [Export] enum RuntimeStatus (L22), [Export] enum DownloadStatus (L30), [Export] interface ProviderManifest (L40), [Export] interface ModelManifest (L53), [Export] interface VoiceManifest (L69), [Export] interface VoiceProfile (L82), [Export] interface TTSRequest (L101), [Export] interface TTSResponse (L117)
app/api/speech/types.ts (cont.)	—	[Export] interface STTRequest (L127), [Export] interface STTResponse (L137), [Export] interface TranscriptionSegment (L146), [Export] interface VoiceChatRequest (L154), [Export] interface VoiceChatResponse (L163), [Export] interface RuntimeConfig (L169), [Export] interface RuntimeHealth (L183), [Export] interface DownloadTask (L195), [Export] interface AudioDevice (L212), [Export] interface BenchmarkResult (L223), [Export] interface BenchmarkConfig (L237), [Export] interface HealthReport (L248), [Export] interface ProviderHealth (L257), [Export] interface ModelHealth (L264), [Export] interface DeviceHealth (L272), [Export] interface IBaseProvider (L280), [Export] interface ITTSProvider (L291), [Export] interface ISTTProvider (L295)
app/api/speech/types.ts (cont.)	—	[Export] interface IHybridProvider (L299), [Export] interface SpeechEventMap (L302), [Export] interface AudioFormat (L328), [Export] interface AudioBuffer (L335), [Export] interface IProviderStorage (L342), [Export] interface IModelStorage (L349), [Export] interface IProfileStorage (L357), [Export] interface IRuntimeStorage (L366), [Export] interface SpeechHistoryEntry (L374), [Export] interface IHistoryStorage (L385), [Export] interface ISettingsStorage (L392), [Export] class SpeechError (L400), [Export] class TypedEventEmitter (L414)
app/api/speech/api/benchmark.ts	43 lines	[Export] const benchmarkRouter (L5)
app/api/speech/api/devices.ts	68 lines	[Export] const devicesRouter (L5)
app/api/speech/api/downloads.ts	154 lines	[Export] const downloadsRouter (L5)
app/api/speech/api/models.ts	92 lines	[Export] const modelsRouter (L5)
app/api/speech/api/profiles.ts	120 lines	[Export] const profilesRouter (L5)
app/api/speech/api/providers.ts	120 lines	[Export] const providersRouter (L5)
app/api/speech/api/runtime.ts	66 lines	[Route] tRPC query: status (L13), [Route] tRPC mutation: stop (L60), [Export] const runtimeRouter (L12)
app/api/speech/api/stt.ts	42 lines	[Export] const sttRouter (L5)
app/api/speech/api/tts.ts	60 lines	[Export] const ttsRouter (L5)
app/api/speech/api/voiceChat.ts	46 lines	[Export] const voiceChatRouter (L5)
app/api/speech/api/voices.ts	120 lines	[Export] const voicesRouter (L5)
app/api/speech/audio/converter.ts	32 lines	[Export] class AudioConverter (L5)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/speech/audio/mp3.ts	69 lines	[Export] interface MP3FrameInfo (L4), [Export] class Mp3Util (L12)
app/api/speech/audio/normalizer.ts	53 lines	[Export] type NormalizationMethod (L4), [Export] class AudioNormalizer (L7)
app/api/speech/audio/player.ts	33 lines	[Export] class AudioPlayer (L5)
app/api/speech/audio/recorder.ts	77 lines	[Export] interface RecorderConfig (L6), [Export] class AudioRecorder (L15)
app/api/speech/audio/resampler.ts	51 lines	[Export] class Resampler (L4)
app/api/speech/audio/silenceDetection.ts	81 lines	[Export] interface SilenceConfig (L4), [Export] interface SilenceResult (L11), [Export] class SilenceDetector (L19)
app/api/speech/audio/stream.ts	32 lines	[Export] class AudioStream (L5)
app/api/speech/audio/wav.ts	92 lines	[Export] class WavUtil (L4)
app/api/speech/events/downloadEvents.ts	27 lines	[Export] fn emitDownloadStarted (L5), [Export] fn emitDownloadProgress (L10), [Export] fn emitDownloadCompleted (L15), [Export] fn emitDownloadCancelled (L20), [Export] fn emitDownloadError (L25)
app/api/speech/events/emitter.ts	21 lines	[Export] class SpeechEventEmitter (L4), [Export] const speechEvents (L21)
app/api/speech/events/providerEvents.ts	22 lines	[Export] fn emitProviderInstalled (L5), [Export] fn emitProviderRemoved (L10), [Export] fn emitProviderEnabled (L15), [Export] fn emitProviderDisabled (L20)
app/api/speech/events/runtimeEvents.ts	17 lines	[Export] fn emitRuntimeStarted (L5), [Export] fn emitRuntimeStopped (L10), [Export] fn emitRuntimeError (L15)
app/api/speech/events/speechEvents.ts	34 lines	[Export] fn emitSpeechGenerated (L5), [Export] fn emitSpeechFailed (L10), [Export] fn emitBenchmarkCompleted (L15), [Export] fn emitBenchmarkFailed (L20), [Export] fn emitTranscriptionCompleted (L25), [Export] fn emitTranscriptionFailed (L30)
app/api/speech/manager/benchmarkManager.ts	69 lines	[Export] class BenchmarkManager (L14)
app/api/speech/manager/deviceManager.ts	110 lines	[Export] class DeviceManager (L10)
app/api/speech/manager/downloadManager.ts	179 lines	[Export] class DownloadManager (L18)
app/api/speech/manager/healthManager.ts	115 lines	[Export] class HealthManager (L15)
app/api/speech/manager/modelManager.ts	144 lines	[Export] class ModelManager (L12)
app/api/speech/manager/profileManager.ts	180 lines	[Export] class ProfileManager (L15)
app/api/speech/manager/providerManager.ts	435 lines	[Export] interface WorkerRef (L54), [Export] interface AggregatedVoiceManifest (L61), [Export] class ProviderManager (L154)
app/api/speech/manager/providerRegistry.ts	100 lines	Internal module / implementation file
app/api/speech/manager/runtimeManager.ts	152 lines	[Export] class RuntimeManager (L23)
app/api/speech/manager/runtimeRegistry.ts	32 lines	[Export] class RuntimeRegistry (L6)
app/api/speech/manager/speechManager.ts	268 lines	[Export] class SpeechManager (L39), [Export] const speechManager (L268)
app/api/speech/profiles/assistant.ts	20 lines	[Export] const assistantProfile (L4)
app/api/speech/profiles/corporate.ts	20 lines	[Export] const corporateProfile (L4)
app/api/speech/profiles/custom.ts	27 lines	[Export] fn createCustomProfile (L4)
app/api/speech/profiles/friendly.ts	20 lines	[Export] const friendlyProfile (L4)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/speech/profiles/index.ts	6 lines	Internal module / implementation file
app/api/speech/profiles/narrator.ts	20 lines	[Export] const narratorProfile (L4)
app/api/speech/profiles/robotic.ts	20 lines	[Export] const roboticProfile (L4)
app/api/speech/providers/index.ts	146 lines	[Export] async fn registerSpeechProviders (L41)
app/api/speech/providers/base/hybridProvider.ts	9 lines	Internal module / implementation file
app/api/speech/providers/base/provider.ts	48 lines	Internal module / implementation file
app/api/speech/providers/base/sttProvider.ts	8 lines	Internal module / implementation file
app/api/speech/providers/base/ttsProvider.ts	8 lines	Internal module / implementation file
app/api/speech/providers/chatterbox/config.ts	20 lines	[Export] interface ChatterboxConfig (L4), [Export] const defaultChatterboxConfig (L13)
app/api/speech/providers/chatterbox/manifest.ts	136 lines	[Export] const chatterboxManifest (L5), [Export] fn getChatterboxManifest (L134)
app/api/speech/providers/chatterbox/provider.ts	208 lines	[Export] class ChatterboxProvider (L24)
app/api/speech/providers/dia/config.ts	20 lines	[Export] interface DiaConfig (L4), [Export] const defaultDiaConfig (L13)
app/api/speech/providers/dia/manifest.ts	124 lines	[Export] fn getDiaManifest (L4)
app/api/speech/providers/dia/provider.ts	133 lines	[Export] class DiaProvider (L21)
app/api/speech/providers/fishspeech/config.ts	20 lines	[Export] interface FishSpeechConfig (L4), [Export] const defaultFishSpeechConfig (L13)
app/api/speech/providers/fishspeech/manifest.ts	106 lines	[Export] const fishSpeechManifest (L5), [Export] fn getFishSpeechManifest (L104)
app/api/speech/providers/fishspeech/provider.ts	132 lines	[Export] class FishSpeechProvider (L21)
app/api/speech/providers/kokoro/config.ts	20 lines	[Export] interface KokoroConfig (L4), [Export] const defaultKokoroConfig (L13)
app/api/speech/providers/kokoro/manifest.ts	151 lines	[Export] const kokoroManifest (L5), [Export] fn getKokoroManifest (L149)
app/api/speech/providers/kokoro/provider.ts	163 lines	[Export] class KokoroProvider (L21)
app/api/speech/providers/piper/config.ts	20 lines	[Export] interface PiperConfig (L4), [Export] const defaultPiperConfig (L13)
app/api/speech/providers/piper/manifest.ts	171 lines	[Export] const piperManifest (L5), [Export] fn getPiperManifest (L169)
app/api/speech/providers/piper/provider.ts	133 lines	[Export] class PiperProvider (L21)
app/api/speech/providers/whisper/config.ts	16 lines	[Export] interface WhisperConfig (L4), [Export] const defaultWhisperConfig (L11)
app/api/speech/providers/whisper/manifest.ts	106 lines	[Export] const whisperManifest (L5), [Export] fn getWhisperManifest (L104)
app/api/speech/providers/whisper/provider.ts	130 lines	[Export] class WhisperProvider (L21)
app/api/speech/providers/xtts/config.ts	22 lines	[Export] interface XTTSConfig (L4), [Export] const defaultXTTSConfig (L14)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/speech/providers/xtts/manifest.ts	106 lines	[Export] const xttsManifest (L5), [Export] fn getXTTSManifest (L104)
app/api/speech/providers/xtts/provider.ts	133 lines	[Export] class XTTSProvider (L21)
app/api/speech/runtimes/clusterHttpClient.ts	191 lines	[Export] class ClusterHttpClient (L28)
app/api/speech/runtimes/config.ts	53 lines	[Export] fn createDefaultRuntimeConfig (L10)
app/api/speech/runtimes/health.ts	37 lines	[Export] class RuntimeHealthChecker (L4)
app/api/speech/runtimes/httpClient.ts	63 lines	[Export] class HttpClient (L5)
app/api/speech/runtimes/logger.ts	40 lines	[Export] interface RuntimeLogEntry (L4), [Export] class RuntimeLogger (L12)
app/api/speech/runtimes/processManager.ts	127 lines	[Export] class ProcessManager (L8)
app/api/speech/runtimes/pythonRuntime.ts	132 lines	[Export] class PythonRuntime (L42)
app/api/speech/runtimes/runtime.ts	58 lines	Internal module / implementation file
app/api/speech/runtimes/runtimeFactory.ts	18 lines	[Export] class RuntimeFactory (L7)
app/api/speech/runtimes/sharedRuntime.ts	88 lines	[Export] class SharedRuntime (L21)
app/api/speech/runtimes/websocket.ts	75 lines	[Export] interface WSMessage (L5), [Export] class WebSocketClient (L11)
app/api/speech/services/clusterSpeech.ts	107 lines	[Export] async fn executeClusterSpeech (L6)
app/api/speech/storage/historyStorage.ts	72 lines	[Export] class HistoryStorage (L10)
app/api/speech/storage/modelStorage.ts	63 lines	[Export] class ModelStorage (L7)
app/api/speech/storage/profileStorage.ts	64 lines	[Export] class ProfileStorage (L6)
app/api/speech/storage/providerStorage.ts	49 lines	[Export] class ProviderStorage (L7)
app/api/speech/storage/runtimeStorage.ts	58 lines	[Export] class RuntimeStorage (L6)
app/api/speech/storage/settingsStorage.ts	51 lines	[Export] class SettingsStorage (L6)
app/api/speech/utils/checksum.ts	30 lines	[Export] fn checksumBuffer (L6), [Export] fn checksumFile (L11), [Export] fn verifyBufferChecksum (L22), [Export] async fn verifyFileChecksum (L27)
app/api/speech/utils/constants.ts	33 lines	[Export] const DEFAULT_SAMPLE_RATE (L2), [Export] const DEFAULT_CHANNELS (L3), [Export] const DEFAULT_BIT_DEPTH (L4), [Export] const DEFAULT_AUDIO_FORMAT (L5), [Export] const DEFAULT_SPEED (L7), [Export] const DEFAULT_PITCH (L8), [Export] const DEFAULT_TEMPERATURE (L9), [Export] const DEFAULT_VOLUME (L10), [Export] const DEFAULT_EMOTION (L11), [Export] const MAX_TEXT_LENGTH (L13), [Export] const MAX_AUDIO_DURATION (L14), [Export] const MAX_DOWNLOAD_RETRIES (L15), [Export] const DOWNLOAD_CHUNK_SIZE (L16), [Export] const DOWNLOAD_TIMEOUT (L17), [Export] const RUNTIME_START_TIMEOUT (L19), [Export] const RUNTIME_STOP_TIMEOUT (L20), [Export] const RUNTIME_HEARTBEAT_INTERVAL (L21), [Export] const RUNTIME_MAX_RETRIES (L22)
app/api/speech/utils/constants.ts (cont.)	—	[Export] const BENCHMARK_WARMUP_ITERATIONS (L24), [Export] const BENCHMARK_DEFAULT_ITERATIONS (L25), [Export] const PROVIDER_IDS (L27), [Export] const SUPPORTED_LANGUAGES (L29), [Export] const AUDIO_FORMATS (L33)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
app/api/speech/utils/downloader.ts	121 lines	[Export] class HttpDownloader (L17)
app/api/speech/utils/fileUtils.ts	102 lines	[Export] async fn ensureDir (L6), [Export] async fn writeFileAtomic (L15), [Export] async fn readJsonFile (L28), [Export] async fn writeJsonFile (L38), [Export] async fn fileExists (L43), [Export] async fn getFileSize (L53), [Export] async fn removePath (L63), [Export] async fn listFiles (L72), [Export] async fn appendFile (L82)
app/api/speech/utils/helpers.ts	79 lines	[Export] fn generateId (L4), [Export] fn generateRequestId (L13), [Export] fn formatBytes (L18), [Export] fn formatDuration (L27), [Export] fn calculateETA (L34), [Export] fn deepClone (L41), [Export] fn isPlainObject (L46), [Export] fn sleep (L51), [Export] async fn withRetry (L56), [Export] fn normalizeProviderId (L76)
app/api/speech/utils/httpClient.ts	102 lines	[Export] interface SpeechHttpRequest (L4), [Export] const speechHttpClient (L101)
app/api/speech/utils/pathUtils.ts	102 lines	[Export] fn getSpeechBasePath (L5), [Export] fn getGeneratedAudioDir (L21), [Export] fn getGeneratedAudioPath (L28), [Export] fn getAudioCacheDir (L35), [Export] fn getAudioCachePath (L39), [Export] fn getAudioTempDir (L46), [Export] fn getAudioTempPath (L50), [Export] fn getProviderPath (L55), [Export] fn getModelPath (L60), [Export] fn getProfileStoragePath (L65), [Export] fn getRuntimeStoragePath (L70), [Export] fn getDownloadTempPath (L75), [Export] fn getSettingsPath (L80), [Export] fn getHistoryPath (L85), [Export] fn sanitizeFilename (L90), [Export] fn validatePathWithinBase (L95)
app/api/speech/utils/validation.ts	104 lines	[Export] fn validateTTSRequest (L13), [Export] fn validateSTTRequest (L27), [Export] fn validateVoiceProfile (L44), [Export] fn sanitizeProfileValues (L79), [Export] fn validateProviderId (L91), [Export] fn validateModelId (L99)
app/db/relations.ts	23 lines	[Export] const messagesRelations (L14), [Export] const messageToolCallsRelations (L18)
app/db/schema.ts	872 lines	[Export] const users (L16), [Export] const folders (L25), [Export] const conversations (L32), [Export] const messages (L45), [Export] const messageToolCalls (L91), [Export] const aiModels (L124), [Export] const systemSnapshots (L140), [Export] const services (L158), [Export] const generatedImages (L168), [Export] const generatedVideos (L257), [Export] const fileFolders (L293), [Export] const files (L300), [Export] const chatAttachments (L311), [Export] const researchCollections (L352), [Export] const researchDocuments (L359), [Export] const knowledgeEntries (L370), [Export] const workflows (L380), [Export] const activityLogs (L390)
app/db/schema.ts (cont.)	—	[Export] const fileContents (L399), [Export] const knowledgeChunks (L427), [Export] const chunkEmbeddings (L463), [Export] const memories (L490), [Export] const memoryChunks (L537), [Export] const memoryChunkEmbeddings (L566), [Export] const chatAttachmentChunks (L591), [Export] const chatAttachmentEmbeddings (L615), [Export] const voiceConversations (L633), [Export] const voiceMessages (L661), [Export] const documentAnalysis (L754), [Export] const forgeSessions (L782), [Export] const forgeTaskNodes (L814), [Export] const forgeIterations (L832), [Export] const forgexSessions (L848), [Export] const forgexOutput (L866)
app/db/seed.ts	490 lines	Internal module / implementation file
app/contracts/errors.ts	15 lines	[Export] const Errors (L7)
app/contracts/types.ts	1 lines	Internal module / implementation file
worker/src/capabilities.ts	78 lines	[Export] async fn getCapabilities (L11)
worker/src/connectionState.ts	40 lines	[Export] const connectionState (L40)
worker/src/executor.ts	91 lines	[Export] interface ExecutionResult (L9), [Export] async fn executeTool (L15)
worker/src/exposureService.ts	662 lines	[Route] GET /internal/status (L319), [Route] POST /internal/toggle (L357), [Route] GET /internal/metrics (L372), [Route] GET /internal/tunnel-url (L391), [Route] GET /ping (L401), [Route] GET /health (L417)
worker/src/jobs.ts	63 lines	[Export] interface WorkerJob (L1), [Export] fn startJob (L15), [Export] fn finishJob (L33), [Export] fn currentJobs (L53), [Export] fn runningJobs (L59)
worker/src/pythonTools.ts	17 lines	[Export] const PYTHON_TOOLS (L1)
worker/src/runtimeStatus.ts	9 lines	[Export] async fn getRuntimeStatus (L2)

Backend TypeScript Path	Lines	Exported Symbols, tRPC Procedures & HTTP Routes (100% Unabridged)
worker/src/server.ts	413 lines	[Route] GET /capabilities (L92), [Route] GET /ping (L114), [Route] GET /logs (L128), [Route] GET /logs/stream (L142), [Route] POST /execute (L179), [Route] POST /analyze-stream (L228), [Route] POST /speech (L230), [Route] POST /tool (L298)
worker/src/speechGateway.ts	199 lines	[Export] interface SpeechHttpRequest (L10), [Export] async fn forwardSpeech (L149)
worker/src/toolGateway.ts	47 lines	[Export] interface ToolHttpRequest (L3), [Export] async fn forwardTool (L15)
worker/src/toolRegistry.ts	28 lines	[Export] const NATIVE_TOOLS (L3), [Export] const ALL_TOOLS (L22)
worker/src/db/index.ts	119 lines	[Export] const db (L15), [Export] async fn initDb (L21), [Export] async fn getConfig (L96), [Export] async fn setConfig (L109)
worker/src/db/schema.ts	61 lines	[Export] const workerConfig (L16), [Export] const accessLog (L23), [Export] const toolHistory (L37), [Export] const metricsSnapshots (L49)
worker/src/runtime/health.ts	67 lines	[Export] async fn waitForHealth (L1)
worker/src/runtime/logBuffer.ts	82 lines	[Export] interface LogLine (L11), [Export] const logBuffer (L69), [Export] fn splitLines (L77)
worker/src/runtime/manager.ts	123 lines	[Export] const runtimeManager (L123)
worker/src/runtime/process.ts	118 lines	[Export] interface RuntimeProcess (L4), [Export] fn startProcess (L14), [Export] fn stopProcess (L104), [Export] fn isRunning (L112)
worker/src/runtime/python.ts	43 lines	[Export] const PYTHON_PORT (L8), [Export] fn startPythonRuntime (L10)
worker/src/runtime/speech.ts	42 lines	[Export] const SPEECH_PORT (L8), [Export] fn startSpeechRuntime (L10)
worker/src/services/analyzeGateway.ts	127 lines	[Export] async fn handleAnalyzeStream (L25)
worker/src/services/python-runtime/client.ts	38 lines	[Export] class PythonRuntimeClient (L3), [Export] const pythonRuntimeClient (L37)
worker/src/services/python-runtime/index.ts	3 lines	Internal module / implementation file
worker/src/services/python-runtime/manager.ts	72 lines	[Export] class PythonRuntimeManager (L5), [Export] const pythonRuntimeManager (L71)
worker/src/services/python-runtime/types.ts	28 lines	[Export] interface PythonRuntimeInfo (L1), [Export] interface PythonExecuteRequest (L19), [Export] interface PythonExecuteResponse (L24)
worker/src/tools/embeddings.ts	0 lines	Internal module / implementation file
worker/src/tools/fileSearch.ts	0 lines	Internal module / implementation file
worker/src/tools/firecrawl.ts	90 lines	[Export] fn formatPageContent (L31)
worker/src/tools/internetSearch.ts	35 lines	[Export] async fn internetSearch (L8)
worker/src/tools/research.ts	81 lines	Internal module / implementation file
worker/src/tools/researchQuery.ts	35 lines	[Export] async fn researchQuery (L4)
worker/src/tools/scientificCalculator.ts	40 lines	[Export] async fn scientificCalculator (L3)
worker/src/tools/semanticSearch.ts	307 lines	[Export] async fn searchKnowledge (L92)
worker/src/tools/tavily.ts	84 lines	[Export] async fn searchInternet (L3)
worker/src/tools/urlReader.ts	43 lines	[Export] async fn urlReader (L4)

4. Complete Python AI Runtimes Symbol & Endpoint Inventory (:8002 & :9000)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/python-runtime/app/__init__.py	0 lines	Module init / configuration constants
app/python-runtime/app/config.py	9 lines	Module init / configuration constants
app/python-runtime/app/health.py	9 lines	def health() (L3)
app/python-runtime/app/main.py	79 lines	def _ensure_port_free() (L13), async def lifespan() (L50)
app/python-runtime/app/runtime.py	155 lines	class Runtime (L35), def __init__() (L37), def info() (L40), async def initialize() (L46), async def shutdown() (L142)
app/python-runtime/app/utils.py	0 lines	Module init / configuration constants
app/python-runtime/app/core/registry.py	16 lines	class Registry (L1), def __init__() (L3), def register() (L6), def get() (L9), def list() (L12), def values() (L15)
app/python-runtime/app/managers/base.py	14 lines	class Manager (L4), async def initialize() (L9), async def shutdown() (L13)
app/python-runtime/app/managers/layout_manager.py	54 lines	class LayoutManager (L7), def __init__() (L11), async def initialize() (L17), def info() (L20), async def register() (L26), def current() (L41), async def shutdown() (L47)
app/python-runtime/app/managers/ocr_manager.py	53 lines	class OCRManager (L6), def __init__() (L10), async def initialize() (L15), def info() (L18), async def register() (L24), def current() (L33), def list() (L37), def set_provider() (L41), async def shutdown() (L48)
app/python-runtime/app/managers/pdf_manager.py	53 lines	class PDFManager (L6), def __init__() (L10), async def initialize() (L15), def info() (L18), async def register() (L24), def current() (L33), def list() (L37), def set_provider() (L41), async def shutdown() (L48)
app/python-runtime/app/managers/runtime_manager.py	20 lines	class RuntimeManager (L1), def __init__() (L3), def register() (L7), def get() (L11), def list() (L15)
app/python-runtime/app/managers/tool_manager.py	47 lines	class ToolManager (L7), def __init__() (L11), async def initialize() (L14), async def register() (L17), def get() (L26), def list() (L30), def info() (L34), async def shutdown() (L41)
app/python-runtime/app/managers/vision_manager.py	57 lines	class VISIONManager (L7), def __init__() (L11), async def initialize() (L16), def models() (L19), def info() (L22), async def register() (L28), def current() (L37), def list() (L41), def set_provider() (L45), async def shutdown() (L52)
app/python-runtime/app/models/ocr.py	6 lines	class OCRRequest (L4)
app/python-runtime/app/models/request.py	8 lines	class ToolRequest (L4)
app/python-runtime/app/models/response.py	10 lines	class ToolResponse (L4)
app/python-runtime/app/processors/layout/parser.py	14 lines	class LayoutParser (L1), def parse() (L4)
app/python-runtime/app/processors/layout/postprocessor.py	25 lines	class LayoutPostProcessor (L1), def cleanup() (L4), def sort_key() (L8)
app/python-runtime/app/processors/layout/preprocessor.py	13 lines	class LayoutPreprocessor (L4), def prepare() (L7)
app/python-runtime/app/processors/ocr/__init__.py	0 lines	Module init / configuration constants
app/python-runtime/app/processors/ocr/parser.py	46 lines	class OCRParser (L1), def parse() (L4)
app/python-runtime/app/processors/ocr/postprocessor.py	14 lines	class OCRPostProcessor (L1), def cleanup() (L4)
app/python-runtime/app/processors/ocr/preprocessor.py	81 lines	class OCRPreprocessor (L4), def load() (L7), def grayscale() (L18), def resize() (L27), def denoise() (L43), def threshold() (L48), def blur() (L60), def sharpen() (L69)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/python-runtime/app/providers/base.py	15 lines	class Provider (L4), async def initialize() (L10), async def shutdown() (L14)
app/python-runtime/app/providers/layout/base.py	10 lines	class LayoutProvider (L6), async def analyze() (L9)
app/python-runtime/app/providers/layout/doclayout.py	0 lines	Module init / configuration constants
app/python-runtime/app/providers/layout/registry.py	8 lines	Module init / configuration constants
app/python-runtime/app/providers/layout/surya.py	47 lines	class SuryaLayoutProvider (L9), def __init__() (L13), async def initialize() (L18), async def analyze() (L30), async def shutdown() (L42)
app/python-runtime/app/providers/ocr/base.py	10 lines	class OCRProvider (L6), async def recognize() (L9)
app/python-runtime/app/providers/ocr/easyocr.py	0 lines	Module init / configuration constants
app/python-runtime/app/providers/ocr/paddle.py	52 lines	class PaddleOCRProvider (L6), def __init__() (L10), async def initialize() (L13), async def recognize() (L24), async def shutdown() (L48)
app/python-runtime/app/providers/ocr/registry.py	9 lines	Module init / configuration constants
app/python-runtime/app/providers/ocr/surya.py	61 lines	class SuryaOCRProvider (L10), def __init__() (L14), async def initialize() (L20), async def recognize() (L38), async def shutdown() (L55)
app/python-runtime/app/providers/ocr/tesseract.py	0 lines	Module init / configuration constants
app/python-runtime/app/providers/pdf/base.py	10 lines	class PDFProvider (L6), async def parse() (L9)
app/python-runtime/app/providers/pdf/marker.py	61 lines	class MarkerPDFProvider (L7), def __init__() (L11), async def initialize() (L16), async def parse() (L32), async def shutdown() (L56)
app/python-runtime/app/providers/pdf/registry.py	7 lines	Module init / configuration constants
app/python-runtime/app/providers/vision/analyze.py	60 lines	class VisionAnalyzeTool (L7), async def initialize() (L11), async def execute() (L15), async def shutdown() (L58)
app/python-runtime/app/providers/vision/base.py	18 lines	class VisionProvider (L8), async def analyze() (L13)
app/python-runtime/app/providers/vision/florence2.py	65 lines	class Florence2Provider (L9), def __init__() (L13), async def initialize() (L19), async def analyze() (L34), async def shutdown() (L62)
app/python-runtime/app/providers/vision/internvl.py	71 lines	class InternVLProvider (L9), def __init__() (L13), async def initialize() (L19), async def analyze() (L43), async def shutdown() (L68)
app/python-runtime/app/providers/vision/minicpm.py	75 lines	class MiniCPMVProvider (L7), def __init__() (L11), async def initialize() (L17), async def analyze() (L21), async def shutdown() (L73)
app/python-runtime/app/providers/vision/models.py	79 lines	Module init / configuration constants
app/python-runtime/app/providers/vision/qwen3vl.py	76 lines	class Qwen3VLProvider (L7), def __init__() (L11), async def initialize() (L17), async def analyze() (L21), async def shutdown() (L74)
app/python-runtime/app/providers/vision/registry.py	17 lines	Module init / configuration constants
app/python-runtime/app/routes/__init__.py	21 lines	Module init / configuration constants
app/python-runtime/app/routes/analyze.py	139 lines	class AnalyzeRequest (L15), def _sse() (L20), def _attribute_failure() (L36), HTTP POST /analyze (L43), async def analyze_stream() (L44), async def event_generator() (L68)
app/python-runtime/app/routes/diagnostics.py	33 lines	HTTP GET /diagnostics (L13), async def diagnostics() (L15)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/python-runtime/app/routes/execute.py	55 lines	async def execute_tool() (L15)
app/python-runtime/app/routes/health.py	10 lines	HTTP GET /health (L8), async def runtime_health() (L9)
app/python-runtime/app/routes/info.py	26 lines	HTTP GET /info (L10), async def info() (L11)
app/python-runtime/app/routes/models.py	50 lines	HTTP GET /models (L11), async def models() (L12)
app/python-runtime/app/routes/ocr.py	22 lines	HTTP POST /ocr (L9), async def run_ocr() (L10)
app/python-runtime/app/routes/providers.py	69 lines	class ProviderRequest (L13), HTTP GET /providers (L18), async def providers() (L19), HTTP POST /providers/{category} (L36), async def switch_provider() (L37)
app/python-runtime/app/routes/runtime.py	36 lines	HTTP GET /runtime (L13), async def runtime_info() (L14)
app/python-runtime/app/tools/base.py	19 lines	class BaseTool (L4), async def initialize() (L10), async def execute() (L14), async def shutdown() (L18)
app/python-runtime/app/tools/document_analysis/__init__.py	5 lines	Module init / configuration constants
app/python-runtime/app/tools/document_analysis/graph.py	277 lines	def _preview() (L35), def _extract_json() (L40), async def _call_ollama_streaming() (L59), async def ingestion_node() (L114), async def summarizer_node() (L124), async def extraction_node() (L138), async def metadata_node() (L152), async def synthesizer_node() (L169), def build_graph() (L195), def get_graph() (L217), async def stream_analysis() (L233), async def on_token() (L246), async def run_graph() (L252)
app/python-runtime/app/tools/document_analysis/prompts.py	88 lines	Module init / configuration constants
app/python-runtime/app/tools/document_analysis/state.py	23 lines	class DocumentAnalysisState (L4)
app/python-runtime/app/tools/document_analysis/tool.py	53 lines	class DocumentAnalysisTool (L11), def __init__() (L15), async def initialize() (L18), async def execute() (L21), async def shutdown() (L52)
app/python-runtime/app/tools/layout/analyzer.py	80 lines	class LayoutAnalyzerTool (L10), async def initialize() (L14), async def execute() (L18), async def shutdown() (L78)
app/python-runtime/app/tools/layout/models.py	6 lines	class LayoutRequest (L4)
app/python-runtime/app/tools/pdf/marker.py	54 lines	class MarkerPDFTool (L7), async def initialize() (L11), async def execute() (L15), async def shutdown() (L52)
app/python-runtime/app/tools/pdf/parser.py	0 lines	Module init / configuration constants
app/python-runtime/app/tools/vision/__init__.py	0 lines	Module init / configuration constants
app/python-runtime/app/tools/vision/analyze.py	75 lines	class VisionAnalyzeTool (L10), async def initialize() (L14), async def execute() (L18), async def shutdown() (L73)
app/python-runtime/app/tools/vision/caption.py	0 lines	Module init / configuration constants
app/python-runtime/app/tools/vision/describe.py	0 lines	Module init / configuration constants
app/python-runtime/app/tools/vision/models.py	25 lines	class VisionObject (L4), class VisionResponse (L13)
app/python-runtime/app/tools/vision/ocr.py	90 lines	class VisionOCRTool (L13), async def initialize() (L17), async def execute() (L21), async def shutdown() (L88)
app/python-runtime/app/tools/vision/question_answer.py	0 lines	Module init / configuration constants
app/python-runtime/app/tools/vision/routes.py	0 lines	Module init / configuration constants

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/speech-runtime/config.py	55 lines	class RuntimeConfig (L10)
app/speech-runtime/install.py	61 lines	def check_python() (L11), def create_directories() (L19), def install_dependencies() (L28), def verify_environment() (L39), def main() (L50)
app/speech-runtime/launcher.py	49 lines	def check_python_version() (L12), def parse_args() (L19), def apply_args() (L30), def launch() (L39)
app/speech-runtime/main.py	40 lines	async def lifespan() (L17), def main() (L26)
app/speech-runtime/api/benchmark.py	34 lines	async def run_benchmark() (L6), async def get_benchmarks() (L31)
app/speech-runtime/api/downloads.py	34 lines	async def start_download() (L6), async def list_downloads() (L21)
app/speech-runtime/api/health.py	9 lines	async def get_health() (L6)
app/speech-runtime/api/models.py	9 lines	async def list_models() (L6)
app/speech-runtime/api/profiles.py	9 lines	async def list_profiles() (L6)
app/speech-runtime/api/providers.py	20 lines	async def list_providers() (L6)
app/speech-runtime/api/stt.py	13 lines	async def transcribe() (L6)
app/speech-runtime/api/tts.py	13 lines	async def synthesize() (L6)
app/speech-runtime/api/voice_chat.py	13 lines	async def voice_chat() (L6)
app/speech-runtime/api/voices.py	34 lines	async def list_voices() (L6)
app/speech-runtime/audio/converter.py	102 lines	class AudioConverter (L11), def convert() (L15), def to_numpy() (L82)
app/speech-runtime/audio/ffmpeg.py	64 lines	class FFmpegUtil (L9), def available() (L12), def decode() (L16)
app/speech-runtime/audio/mp3.py	46 lines	class Mp3Util (L8), def decode() (L12), def encode() (L31)
app/speech-runtime/audio/normalizer.py	36 lines	class AudioNormalizer (L7), def normalize_peak() (L11), def normalize_rms() (L20), def normalize() (L29)
app/speech-runtime/audio/player.py	46 lines	class AudioPlayer (L11), def __init__() (L14), def set_callback() (L18), async def play() (L22), async def stop() (L39), def is_playing() (L44)
app/speech-runtime/audio/recorder.py	66 lines	class RecordingConfig (L13), class AudioRecorder (L22), def __init__() (L25), def set_callback() (L31), async def start() (L35), async def stop() (L43), def feed() (L53), def is_recording() (L64)
app/speech-runtime/audio/resampler.py	32 lines	class Resampler (L7), def resample() (L11), def resample_pcm() (L24)
app/speech-runtime/audio/silence.py	65 lines	class SilenceResult (L10), class SilenceDetector (L18), def __init__() (L21), def detect() (L26)
app/speech-runtime/audio/stream.py	48 lines	class AudioStreamBuffer (L11), def __init__() (L14), async def put() (L19), async def get() (L24), async def iter_chunks() (L33), def close() (L42), def __aiter__() (L47)
app/speech-runtime/audio/vad.py	54 lines	class VADSegment (L10), class VoiceActivityDetector (L17), def __init__() (L20), def detect() (L25)
app/speech-runtime/audio/wav.py	85 lines	class WavUtil (L10), def parse_header() (L14), def extract_pcm() (L26), def build_wav() (L39), def pcm_to_numpy() (L64), def numpy_to_pcm() (L76)
app/speech-runtime/inference/batching.py	75 lines	class BatchTTSItem (L13), class BatchSTTItem (L21), class BatchTTS (L28), def __init__() (L31), async def process() (L34), class BatchSTT (L53), def __init__() (L56), async def process() (L59)
app/speech-runtime/inference/streaming.py	65 lines	class StreamingTTS (L13), def __init__() (L16), async def stream() (L20), class StreamingSTT (L38), def __init__() (L41), async def feed_audio() (L45), async def transcribe_stream() (L49)
app/speech-runtime/inference/stt.py	73 lines	class STTInference (L15), def __init__() (L18), async def transcribe() (L22)
app/speech-runtime/inference/tts.py	75 lines	class TTSInference (L16), def __init__() (L19), async def synthesize() (L24)
app/speech-runtime/inference/voice_chat.py	47 lines	class VoiceChatInference (L13), def __init__() (L16), async def process() (L20)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/speech-runtime/models/cache.py	55 lines	class ModelCache (L11), def __init__() (L14), async def get() (L19), async def put() (L27), async def remove() (L40), async def clear() (L47), def keys() (L53)
app/speech-runtime/models/downloader.py	121 lines	class DownloadTask (L19), class ModelDownloader (L35), def __init__() (L38), async def download() (L43), def progress_callback() (L72), def get_task() (L102), def list_active() (L106), async def cancel() (L110), async def close() (L119)
app/speech-runtime/models/loader.py	55 lines	class ModelLoader (L13), def __init__() (L16), async def load() (L20), async def unload() (L33), def is_loaded() (L41), def get() (L46), async def clear() (L51)
app/speech-runtime/models/manager.py	99 lines	class ModelManager (L18), def __init__() (L21), async def register_model() (L28), async def get_model() (L36), async def list_models() (L40), async def list_by_provider() (L44), async def delete_model() (L48), async def download_model() (L58), async def load_model() (L72), async def unload_model() (L82), async def is_loaded() (L87), async def get_loaded_models() (L91), async def shutdown() (L95)
app/speech-runtime/models/validator.py	37 lines	class ModelValidator (L12), def validate_file() (L16), def validate_metadata() (L30)
app/speech-runtime/providers/base.py	322 lines	class ProviderManifest (L9), def __init__() (L33), def type() (L76), class ModelInfo (L82), class VoiceInfo (L101), class TTSRequest (L114), class TTSResponse (L139), class STTRequest (L150), class STTResponse (L161), class BenchmarkConfig (L171), class BenchmarkResult (L183), class ProviderStats (L193), class BaseProvider (L202), def __init__() (L205), def id() (L215), def manifest() (L221), def is_initialized() (L226), async def initialize() (L231)
app/speech-runtime/providers/base.py (cont.)	—	async def shutdown() (L236), async def health() (L241), async def list_models() (L246), async def get_model() (L250), async def list_voices() (L259), async def get_voice() (L263), async def install_model() (L271), async def install_voice() (L275), async def uninstall_voice() (L279), async def load_model() (L284), async def unload_model() (L289), async def synthesize() (L294), async def transcribe() (L299), async def stats() (L303), async def benchmark() (L315), def _ensure_initialized() (L319)
app/speech-runtime/providers/chatterbox.py	505 lines	class ChatterboxProvider (L58), def __init__() (L62), def id() (L77), def manifest() (L81), async def initialize() (L105), async def shutdown() (L159), async def health() (L165), async def list_models() (L182), async def list_voices() (L228), async def load_model() (L296), async def unload_model() (L306), async def synthesize() (L320), async def transcribe() (L441), async def benchmark() (L448)
app/speech-runtime/providers/dia.py	207 lines	class DiaProvider (L23), def id() (L30), def manifest() (L34), async def initialize() (L47), async def shutdown() (L53), async def health() (L61), async def list_models() (L69), async def list_voices() (L82), def _load_model() (L90), async def load_model() (L120), async def unload_model() (L125), def _build_dia_text() (L132), async def synthesize() (L143), async def transcribe() (L195), async def benchmark() (L198)
app/speech-runtime/providers/fishspeech.py	346 lines	class FishSpeechProvider (L32), def __init__() (L34), def id() (L51), def manifest() (L55), async def initialize() (L85), async def shutdown() (L106), async def health() (L112), async def list_models() (L129), async def list_voices() (L156), async def load_model() (L197), async def unload_model() (L201), async def synthesize() (L205), async def transcribe() (L317), async def benchmark() (L327)
app/speech-runtime/providers/kokoro.py	296 lines	def _normalize_text_for_tts() (L59), class KokoroProvider (L108), def __init__() (L112), def id() (L125), def manifest() (L129), async def initialize() (L147), async def shutdown() (L187), async def health() (L191), async def list_models() (L202), async def list_voices() (L215), async def load_model() (L226), async def unload_model() (L232), async def synthesize() (L235), async def transcribe() (L284), async def benchmark() (L287)
app/speech-runtime/providers/piper.py	178 lines	class PiperProvider (L37), def id() (L44), def manifest() (L48), async def initialize() (L61), async def shutdown() (L67), async def health() (L73), async def list_models() (L80), async def list_voices() (L88), def _get_model_path() (L100), def _load_voice() (L107), async def load_model() (L126), async def unload_model() (L129), async def synthesize() (L134), async def transcribe() (L166), async def benchmark() (L169)
app/speech-runtime/providers/registry.py	110 lines	class ProviderRegistry (L17), def __init__() (L20), def discover() (L26), async def register() (L58), async def unregister() (L69), async def reload() (L78), def get() (L86), def list_all() (L90), def list_ids() (L94), async def initialize_all() (L98), async def shutdown_all() (L107)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/speech-runtime/providers/whisper.py	141 lines	class WhisperProvider (L21), def id() (L25), def manifest() (L29), async def initialize() (L42), async def shutdown() (L48), async def health() (L53), async def list_models() (L56), async def list_voices() (L62), async def load_model() (L65), async def unload_model() (L93), async def synthesize() (L97), async def transcribe() (L100), async def benchmark() (L138)
app/speech-runtime/providers/xtts.py	188 lines	class XTTSProvider (L27), def id() (L34), def manifest() (L38), async def initialize() (L51), async def shutdown() (L57), async def health() (L65), async def list_models() (L73), async def list_voices() (L79), def _load_tts() (L88), async def load_model() (L106), async def unload_model() (L111), async def synthesize() (L118), async def transcribe() (L176), async def benchmark() (L179)
app/speech-runtime/runtime/app.py	44 lines	def create_app() (L15), async def default_lifespan() (L32)
app/speech-runtime/runtime/benchmark.py	127 lines	class BenchmarkResult (L16), def to_dict() (L31), class BenchmarkRunner (L35), def __init__() (L38), async def run_tts_benchmark() (L42), async def run_stt_benchmark() (L88)
app/speech-runtime/runtime/config.py	70 lines	class RuntimeServerConfig (L6), class AudioConfig (L16), class InferenceConfig (L24), class CacheConfig (L32), class WebSocketConfig (L39), class EngineConfig (L47), class RuntimeConfig (L62)
app/speech-runtime/runtime/engine_manager.py	146 lines	class EngineManager (L16), def __init__() (L19), def _engine_dir() (L26), def _python() (L29), def _launcher() (L37), def _port() (L40), def base_url() (L43), def get_url() (L46), def is_running() (L49), def health() (L57), def start() (L69), def stop() (L116), def restart() (L132), def get_engine_manager() (L140)
app/speech-runtime/runtime/events.py	69 lines	class Event (L13), class EventBus (L23), def __init__() (L26), def subscribe() (L31), def unsubscribe() (L38), async def emit() (L43), async def _safe_call() (L54), def get_history() (L61)
app/speech-runtime/runtime/exceptions.py	59 lines	class SpeechRuntimeError (L4), def __init__() (L6), class ProviderError (L12), def __init__() (L14), class ModelError (L19), def __init__() (L21), class InferenceError (L26), def __init__() (L28), class AudioError (L32), def __init__() (L34), class ValidationError (L38), def __init__() (L40), class DownloadError (L44), def __init__() (L46), class NotFoundError (L50), def __init__() (L52), class NotSupportedError (L56), def __init__() (L58)
app/speech-runtime/runtime/health.py	124 lines	class ProviderHealth (L15), class ModelHealth (L24), class DeviceHealth (L34), class HealthReport (L43), def to_dict() (L58), class HealthMonitor (L63), def __init__() (L66), def add_error() (L71), def add_warning() (L76), def get_report() (L81), def clear() (L121)
app/speech-runtime/runtime/logger.py	43 lines	def setup_logging() (L16), def get_logger() (L39)
app/speech-runtime/runtime/provider_loader.py	96 lines	class ProviderLoader (L13), def __init__() (L16), def _discover_and_index() (L25), def get_class() (L48), async def load_inprocess() (L57), async def load_all() (L83), async def reload() (L94)
app/speech-runtime/runtime/registration.py	124 lines	class ProviderRegistration (L35), class HeartbeatPayload (L44), class RegistrationManager (L55), def __init__() (L58), def record_heartbeat() (L68), def untrack() (L73), def start() (L77), def stop() (L83), async def _watchdog() (L94)
app/speech-runtime/runtime/request_handler.py	75 lines	class RequestHandler (L17), def __init__() (L19), async def handle_tts() (L22), async def handle_stt() (L40), async def handle_voice_chat() (L59)
app/speech-runtime/runtime/router.py	284 lines	def setup_routes() (L12), HTTP GET /health (L15), async def health() (L16), HTTP GET /providers (L27), async def list_providers() (L28), HTTP GET /models (L44), async def list_models() (L45), HTTP GET /voices (L50), async def list_voices() (L51), HTTP GET /profiles (L94), async def list_profiles() (L95), HTTP POST /tts (L100), async def tts() (L101), HTTP POST /stt (L110), async def stt() (L111), HTTP POST /voice-chat (L120), async def voice_chat() (L121), HTTP POST /download (L130)
app/speech-runtime/runtime/router.py (cont.)	—	async def download() (L131), HTTP DELETE /model/{model_id} (L145), async def delete_model() (L146), HTTP GET /benchmark (L151), async def get_benchmarks() (L152), HTTP POST /benchmark (L157), async def run_benchmark() (L158), HTTP GET /events (L182), async def get_events() (L183), HTTP GET /logs (L198), async def get_logs() (L199), HTTP GET /downloads (L205), async def list_downloads() (L206), HTTP POST /register (L225), async def register_engine() (L226), HTTP POST /heartbeat (L257), async def engine_heartbeat() (L258), async def speech_runtime_exception_handler() (L279)

Python Runtime File Path	Lines	FastAPI Routes, Classes & Functions (100% Unabridged)
app/speech-runtime/runtime/runtime_manager.py	283 lines	class RuntimeManager (L28), def __init__() (L31), async def initialize() (L96), async def shutdown() (L158), async def register_provider() (L204), async def handle_heartbeat() (L244), async def get_health() (L253)
app/speech-runtime/runtime/voice_library.py	111 lines	class VoiceLibrary (L6), def __init__() (L8), def load_all() (L20), def get() (L105), def list() (L109)
app/speech-runtime/runtime/websocket_manager.py	75 lines	class WebSocketManager (L15), def __init__() (L18), async def connect() (L22), async def disconnect() (L29), async def broadcast() (L35), async def handle_stream() (L46), async def subscribe_events() (L66), async def event_handler() (L68)
app/speech-runtime/storage/history.py	46 lines	class HistoryStorage (L9), def __init__() (L12), def append() (L16), def load_all() (L23), def load_recent() (L38), def clear() (L43)
app/speech-runtime/storage/models.py	47 lines	class ModelStorage (L8), def __init__() (L11), def _file() (L14), def save() (L17), def load() (L22), def delete() (L30), def list_all() (L36), def list_by_provider() (L45)
app/speech-runtime/storage/profiles.py	51 lines	class ProfileStorage (L8), def __init__() (L11), def _file() (L14), def save() (L17), def load() (L22), def delete() (L30), def list_all() (L36), def get_default() (L45)
app/speech-runtime/storage/providers.py	43 lines	class ProviderStorage (L8), def __init__() (L11), def _file() (L14), def save() (L17), def load() (L22), def delete() (L30), def list_all() (L36)
app/speech-runtime/storage/settings.py	49 lines	class SettingsStorage (L10), def __init__() (L13), def _load() (L19), def _save() (L27), def get() (L32), def set() (L36), def delete() (L41), def all() (L47)
app/speech-runtime/utils/checksum.py	32 lines	def checksum_buffer() (L9), def checksum_file() (L16), def verify_buffer() (L25), def verify_file() (L30)
app/speech-runtime/utils/constants.py	37 lines	Module init / configuration constants
app/speech-runtime/utils/downloader.py	92 lines	class DownloadProgress (L16), class HttpDownloader (L24), def __init__() (L27), async def _get_session() (L31), async def download() (L37), def cancel() (L85), async def close() (L89)
app/speech-runtime/utils/environment.py	51 lines	def get_python_version() (L8), def get_system_info() (L13), def get_gpu_info() (L26), def get_device_preference() (L43)
app/speech-runtime/utils/helpers.py	65 lines	def generate_id() (L8), def generate_request_id() (L13), def format_bytes() (L18), def format_duration() (L30), def calculate_eta() (L37), def deep_clone() (L45), def is_plain_object() (L51), def current_timestamp() (L56), def sanitize_filename() (L61)
app/speech-runtime/utils/paths.py	46 lines	def get_base_path() (L8), def get_cache_path() (L13), def get_logs_path() (L18), def get_output_path() (L23), def get_temp_path() (L28), def get_models_path() (L33), def get_voices_path() (L38), def ensure_dir() (L43)
app/speech-runtime/utils/timer.py	50 lines	class TimingResult (L10), class Timer (L17), def __init__() (L20), def __enter__() (L24), def __exit__() (L28), def elapsed() (L32), def elapsed_ms() (L37), async def async_timer() (L43)
app/speech-runtime/utils/validation.py	54 lines	def validate_tts_request() (L7), def validate_stt_request() (L16), def validate_language() (L31), def validate_profile() (L37)
app/speech-runtime/voices/loader.py	46 lines	class VoiceLoader (L13), def __init__() (L16), async def load() (L20), async def unload() (L31), def is_loaded() (L39), def get_voice_path() (L44)
app/speech-runtime/voices/profiles.py	151 lines	class VoiceProfile (L17), def to_dict() (L35), def from_dict() (L55), class ProfileManager (L75), def __init__() (L78), def register_builtin() (L82), async def initialize() (L87), async def create_profile() (L95), async def get_profile() (L110), async def get_default() (L117), async def list_profiles() (L124), async def update_profile() (L129), async def delete_profile() (L143)
app/speech-runtime/voices/registry.py	41 lines	class VoiceRegistry (L10), def __init__() (L13), def register() (L16), def unregister() (L21), def get() (L27), def list_all() (L31), def list_by_provider() (L35), def clear() (L39)
app/speech-runtime/voices/validator.py	33 lines	class VoiceValidator (L7), def validate_speed() (L11), def validate_pitch() (L16), def validate_temperature() (L21), def validate_volume() (L26), def validate_voice_id() (L31)